SEASUG Paper 2026 year

Predicting Viewership and Tracking Audience Reactions to AI-Generated Music on YouTube Over Time: A Predictive Modeling and LLM-Assisted Approach

Mohsen Pashna, Oklahoma State University; Hunter Willard, Oklahoma State University

ABSTRACT

This study examines AI-generated music on YouTube from two connected perspectives: what characteristics are associated with high viewership and how audience reactions to AI-generated music change over time. For the first research question, 3,967 AI-generated music videos were analyzed using logistic regression, decision tree, random forest, and XGBoost models, followed by feature-importance and genre analyses. XGBoost provided the strongest overall predictive performance, and the results showed that content-related characteristics, especially genre, artist-mimicry or cover references, and AI-vocal references, were more important for predicting high viewership than basic metadata such as hashtags, description length, or mentions of specific AI platforms. For the second research question, 46,443 multilingual YouTube comments from 2,344 videos were analyzed using LLM-assisted theme discovery, translation, sentiment coding, and quarterly trend analysis. The results show that audience discussion gradually moved away from early fascination with realism and voice recreation toward greater attention to artist impact, ethics, consent, credit, and limitations of AI-generated music, while overall positive sentiment also declined over time. Taken together, the findings suggest that AI-generated music is moving beyond a stage of technological novelty. As technology becomes more common, success appears to depend more on the type and positioning of the music itself, while audiences increasingly judge AI-generated content based on its quality, fairness, and effects on human artists. The study therefore provides an integrated view of both the platform performance and changing audience reception of AI-generated music.

Keywords: AI-generated music; YouTube; music popularity; audience reception; machine learning; XGBoost; sentiment analysis; thematic analysis; generative AI; social media analytics

INTRODUCTION

Recent advances in generative artificial intelligence have rapidly expanded the possibilities for music creation and distribution. Text-to-music systems such as MusicLM and MusicGen, together with commercial platforms capable of generating complete songs with vocals, have moved AI-generated music from experimental research settings into widespread public use (Agostinelli et al., 2023; Copet et al., 2023). AI-generated songs, synthetic vocals, artist-style imitations, and AI-assisted covers now circulate widely on digital platforms such as YouTube, where they can attract substantial audiences and generate extensive public discussion. This development creates a new environment in which AI-generated music can be examined not only as a technological innovation, but also as a form of digital content competing for audience attention and as a cultural product that listeners actively interpret and evaluate.

Research on music popularity has traditionally focused on identifying the characteristics associated with successful songs. Prior studies have shown that acoustic characteristics, metadata, and platform-level information can contribute to predicting music popularity (Lee & Lee, 2018; Martín-Gutiérrez et al., 2020; Saragih, 2023). Other research has expanded this perspective by incorporating social and platform-related information, including YouTube measures, into popularity-prediction models. However, this literature has largely focused on conventional music, streaming platforms, or general measures of music popularity rather than the emerging context of AI-generated music on YouTube. AI-generated music introduces additional characteristics that are largely absent from conventional popularity models, including explicit AI disclosure, references to generative platforms such as Suno and Udio, AI-generated vocals, and artist mimicry or cover content. These characteristics raise an important question about whether the factors associated with the visibility of AI-generated music differ from those previously identified for conventional music.

At the same time, understanding the development of AI-generated music requires attention not only to which content achieves greater visibility, but also to how audiences respond to it. YouTube comments offer a valuable source for examining these responses because they are unsolicited, time-stamped, multilingual, and directly connected to the content being viewed. Existing research has examined audience responses to online music performances, adoption intentions toward generative AI for music composition, and perceptions of AI-generated and human–AI co-created music (Fraser et al., 2021; Zhou et al., 2025; Rigole et al., 2025). These studies provide useful evidence about listener attitudes and acceptance; however, much of the existing work is cross-sectional or based on surveys and relatively small samples. Such approaches provide snapshots of audience perceptions but are less suited to understanding how public discourse changes as AI-music technologies become more familiar and as questions involving authenticity, artists' livelihoods, consent, copyright, disclosure, and creative ownership become increasingly visible.

A longitudinal perspective is particularly important because both AI-music technologies and their surrounding social and industry environments have changed rapidly. Early audience responses may be shaped strongly by the novelty of hearing AI reproduce human-like music or imitate recognizable performers, whereas later responses may increasingly reflect practical experience with generative tools as well as concerns about quality, artist impact, consent, attribution, and copyright. The existing RQ2 evidence suggests precisely this type of transition: early discussion was heavily centered on realism and voice recreation, while later discourse placed greater emphasis on artist impact, ethics, credit, and technological limitations. Consequently, treating audience perceptions of AI-generated music as static may overlook meaningful changes in how listeners understand and evaluate the technology over time.

The performance and reception perspectives therefore represent complementary dimensions of the same emerging AI-music ecosystem. Video-level analysis can identify which characteristics are most useful for distinguishing highly viewed AI-generated music from lower-viewership content, while audience-level analysis can reveal how listeners interpret that content and how those interpretations evolve. Considering only one dimension would provide an incomplete picture. Characteristics such as musical genre, AI-generated vocals, artist mimicry, or disclosure may influence the visibility and positioning of AI-generated music, while related characteristics may simultaneously become the focus of audience discussions about realism, authenticity, creativity, artist consequences, and ethical concerns. Examining these dimensions within a single research project therefore provides a broader perspective on both the performance of AI-generated music and its evolving public reception.

Accordingly, this study examines AI-generated music on YouTube through two complementary but analytically distinct research questions. The first focuses on the characteristics associated with high video viewership, while the second examines how audience discussion surrounding AI-generated music changes over time. The first analysis uses a video-level dataset containing 3,967 unique AI-generated music videos, whereas the second draws on 46,443 public YouTube comments from 2,344 videos, covering the period from July 2023 through May 2026. The research questions are as follows:

RQ1: What content, metadata, and AI-related characteristics are most important in predicting whether AI-generated music videos achieve high viewership on YouTube?

RQ2: How do the dominant themes in YouTube comments about AI-generated music change over time?

Following this introduction, the paper is divided into three sections. RQ1 addresses the first research question and presents the relevant literature review, data and methodology, analysis, and findings concerning the characteristics associated with high viewership of AI-generated music videos. RQ2 addresses the second research question its literature review, data and methodology, and longitudinal analysis of audience themes and sentiment. Conclusion then brings the two research questions together through an integrated overall discussion of their findings, followed by the study's limitations and directions for future research.

RQ1: PREDICTORS OF HIGH VIEWERSHIP OF AI-GENERATED MUSIC

LITERATURE REVIEW

Prior research on music-popularity prediction has examined a broad range of acoustic, metadata, and platform-based indicators. Lee and Lee (2018), for example, demonstrated that acoustic characteristics could provide meaningful information for distinguishing popular music, while Martín-Gutiérrez et al. (2020) combined audio representations and metadata within a deep-learning framework for popularity prediction. More recent studies have expanded this perspective by applying ensemble learning methods and incorporating information from digital platforms. Yee and Raheem (2022), for instance, combined Spotify characteristics with YouTube-related measures and found that social-platform information improved predictive performance. Collectively, these studies demonstrate that music popularity is shaped by multiple dimensions rather than by acoustic characteristics alone. Nevertheless, existing research provides limited evidence about the specific characteristics associated with the popularity of AI-generated music, particularly on YouTube. AI-generated music introduces features that are largely absent from conventional popularity models, including disclosure of AI involvement, references to generative tools, AI-generated vocals, and artist-mimicry or cover signals. The present study therefore extends prior music-popularity research by examining these AI-specific characteristics together with genre, publication timing, textual metadata, tags, hashtags, duration, and video age. In addition, rather than allowing established channel size to dominate prediction, the content-focused analysis excludes raw channel-scale variables and combines machine-learning prediction with interpretable genre-level logistic-regression estimates.

DATA AND METHODOLOGY

Data and Analytical Sample

The analysis used a video-level dataset of AI-generated music content collected from YouTube. The final analytical sample for RQ1 contained 3,967 unique videos. Each observation represented a single YouTube video and included information describing the creator/channel, publication timing, music genre, video characteristics, metadata, AI-related characteristics, disclosure, and observed viewership.

The initial predictor set included channel subscriber count, total channel views, channel video count, channel age, subscriber category, publication hour and day of week, days since upload, genre, video category, duration, title and description characteristics, tag and hashtag counts, mentions of AI-music platforms such as Suno and Udio, indicators for AI vocals and artist mimicry or covers, and AI disclosure strength. The raw view_count variable was reserved for constructing the dependent variable and was excluded from the predictor matrix. Figure 1 represents the data pipeline for the analysis of RQ1.

Dependent Variable

RQ1 examines the characteristics associated with high-viewership AI-generated music videos. A binary dependent variable, is_high_engagement, was constructed from each video's view count. Videos with more than the 75th percentile of views (4,058 views) were classified as high engagement (1), while videos with 4,058 or fewer views were classified as lower engagement (0). This procedure produced 992 high-engagement videos (25.01%) and 2,975 lower-engagement videos (74.99%). Using the upper quartile provided a clearly defined classification target representing videos that achieved relatively high viewership within the observed sample.

Data Preparation and Feature Engineering

Several preprocessing procedures were applied before model estimation. Publication timing was represented through the video's UTC publication hour and day of week. Categorical predictors, including music genre and subscriber categories, were encoded for models requiring numerical inputs. Missing numerical values were imputed using the median, while missing categorical observations were assigned a separate "missing" category. Numerical predictors were standardized, and categorical predictors were one-hot encoded, with the first category omitted to avoid redundant dummy variables. A list of predictor variables used in the models are shown in Table 3.To prevent data leakage and multicollinearity, raw channel-scale metrics, such as subscriber counts, total views, and total videos, were excluded. The models instead focused on content-focused and metadata-focused features.

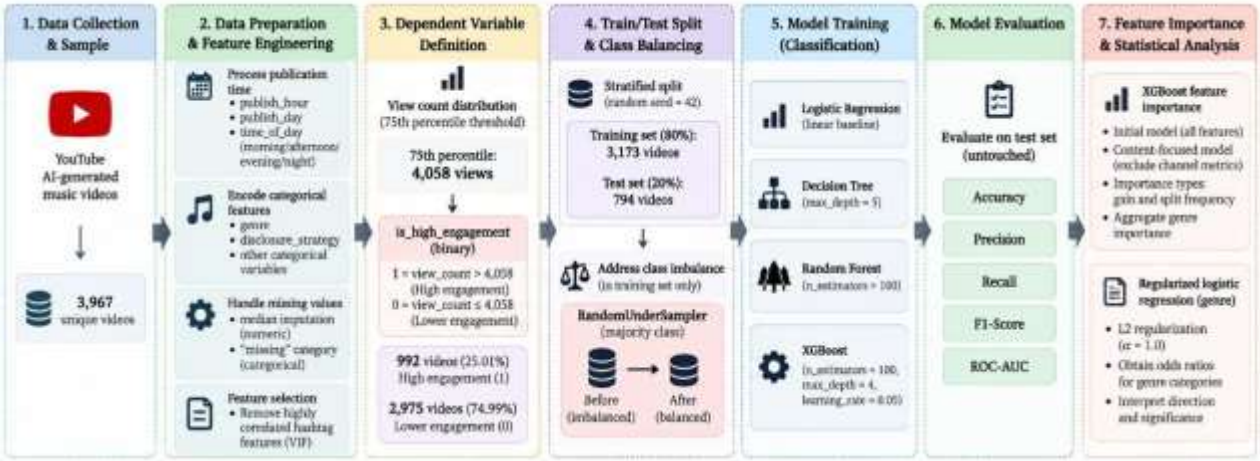


Figure 1. Analysis Pipeline for RQ1

The data were divided into 80% training and 20% testing samples using a stratified split to preserve the approximately 75/25 distribution of the dependent variable in both sets. A fixed random seed of 42 was used for reproducibility. This resulted in 3,173 training observations and 794 test observations. Because high-engagement videos represented only approximately one-quarter of the sample, class imbalance was addressed during the initial model comparison using random undersampling of the majority class within the training pipeline. Importantly, resampling was performed only during model training, leaving the test data unchanged for evaluation. Multicollinearity was also examined using variance inflation factors (VIFs). Highly overlapping hashtag-count measures were identified during this assessment and were excluded from the subsequent statistical specification where appropriate to improve coefficient stability.

Predictive Models and Model Evaluation

Four supervised classification algorithms were evaluated: logistic regression, decision tree, random forest, and XGBoost. Logistic regression provided a linear benchmark, while the tree-based methods allowed nonlinear relationships and interactions among video characteristics.The decision tree was restricted to a maximum depth of five, and the random forest contained 100 trees. The initial XGBoost specification used 100 boosting iterations, a maximum tree depth of four, and a learning rate of 0.05. All models were trained using the same stratified training/test framework and random seed.Out-of-sample performance was evaluated on the untouched test set using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (ROC-AUC). Because the outcome was imbalanced, particular attention was given to F1-score and ROC-AUC rather than relying on accuracy alone. The best-performing model was subsequently used for more detailed feature-importance analysis.

Feature Importance and Statistical Interpretation

Following the model comparison, XGBoost was used to examine which characteristics contributed most strongly to high-engagement classification. Initial analysis indicated that channel-scale variables—particularly subscriber count and total channel views—dominated predictive importance. Because the substantive objective of RQ1 was to identify characteristics of AI-generated music content rather than simply reproduce differences in pre-existing channel size, a secondary content-focused specification excluded channel subscriber count, total channel views, total channel video count, channel age, and subscriber-size category.

Study (Year)	Data & Sample	Methods/Models	Key Findings	Novelty/Contribution
Lee & Lee (2018)	Long-term chart/streaming data (pop songs, varied genres)	Acoustic analysis (MFCCs, complexity); classification/regression	Audio features (complexity, MFCC) predict song popularity metrics significantly above chance.	Pioneered defining popularity metrics and using audio content for prediction.
Martín-Gutiérrez et al. (2020)	Spotify recordings (≈2,000 tracks from top 100 artists)	Deep CNN ("HitMusicNet") on spectrogram + metadata	Multimodal model achieved high accuracy (~97% F1) in predicting track popularity (Spotify metrics).	First end-to-end neural architecture combining audio and metadata for popularity.
Saragih (2023)	92,755 Spotify tracks (Kaggle dataset, Indonesia)	Machine learning (Extra Trees, Random Forest, logistic regression)	Tree ensembles (Extra Trees, Random Forest) achieved highest accuracy; audio features significantly influence popularity.	Large-scale analysis of Spotify audio features in hit-song prediction for a non-Western market.
Yee & Raheem (2022)	Newly released tracks (May–Aug 2021); Spotify+YouTube data	ML classification (Random Forest, etc.) on audio and social media features	Random forest yielded ~79.6% accuracy; including YouTube social metrics (views, likes, etc.) boosted performance 10–60%.	Integrated audio features with YouTube social data to predict hit potential (first such study).
This Study (2026)	3,967 AI-generated YouTube music videos	Content-focused XGBoost + logistic regression	Content signals (genre, AI-vocal mentions, cover labels) strongly predict high views; many genres (e.g., Lo-Fi, EDM) have 40–70% lower odds of hit compared to classical/jazz ($p<0.05$). Metadata features (hashtags, titles) add little predictive power.	First comprehensive study of AI-generated music on YouTube; combines interpretable models to isolate content signals (genre, AI-tool cues) as drivers of viewership.

Table 1. Comparison of Prior Music Popularity Prediction Studies and the Present Study

Feature Category	Predictor	Type / Coding	Description / Model Treatment
Content-Focused Features	genre	Categorical	Classification of the video's musical genre, including Multi-Genre, Lo-Fi/Ambient, EDM, Pop, R&B, and others. In logistic regression, genre was converted into dummy variables with **Classical / Jazz / Soundtracks / Instrumental** as the reference category.
Content-Focused Features	mentions_cover_or_mimic	Binary (1/0)	Indicates whether the video is presented as an artist voice imitation or song cover.
Content-Focused Features	mentions_ai_vocals	Binary (1/0)	Indicates whether the video metadata specifically references AI vocal tools.
Content-Focused Features	mentions_suno	Binary (1/0)	Indicates whether Suno was explicitly referenced in the video metadata.
Content-Focused Features	mentions_udio	Binary (1/0)	Indicates whether Udio was explicitly referenced in the video metadata.
Content-Focused Features	disclosure_strength	Ordinal integer	Measures the degree to which the creator transparently disclosed the use of synthetic AI tools in the video title or description.
Video Metadata & Timeline Features	days_since_upload	Continuous	Number of days the video has been active on YouTube, capturing the potential effect of video age and engagement decay.
Video Metadata & Timeline Features	duration_seconds	Continuous	Total length of the music video in seconds.
Video Metadata & Timeline Features	description_length_chars	Continuous	Number of characters contained in the video description.
Video Metadata & Timeline Features	title_word_count	Continuous	Number of words contained in the video title.
Video Metadata & Timeline Features	title_length_chars	Continuous	Number of characters contained in the video title.
Video Metadata & Timeline Features	tag_count	Continuous	Total number of search tags associated with the video.
Video Metadata & Timeline Features	published_hour_utc	Integer (0–23)	UTC hour at which the video was published.
Video Metadata & Timeline Features	published_day_of_week_utc	Integer (0–6)	Day of the week on which the video was published.
Hashtag Features	hashtags_title_count	Count	Number of hashtags appearing in the video title.
Hashtag Features	hashtags_description_count	Count	Number of hashtags appearing in the video description.
Hashtag Features	hashtags_total_count	Count	Total number of hashtags associated with the video. VIF analysis indicated very low multicollinearity with the remaining predictors, supporting its inclusion in the logistic regression model.

Table 2. Predictor Variables (Features) Used in Models

The refined XGBoost model used 150 estimators, a maximum depth of six, a learning rate of 0.05, and 80% feature subsampling per tree. Class imbalance was addressed through XGBoost's class-weighting mechanism (scale_pos_weight) rather than discarding majority-class observations. Feature importance was examined using both split frequency and gain. Because genre was represented by multiple categories after one-hot encoding in the final gain-based specification, the importance contributions of individual genre indicators were aggregated to produce a single conceptual measure of genre importance.

Finally, a complementary logistic regression analysis was conducted to provide interpretable odds ratios for individual music genres. The initial maximum-likelihood specification exhibited convergence problems associated with multicollinearity and quasi-separation. Rather than removing multiple sparse genre categories and losing observations, the final model applied stronger L2 regularization ($\alpha = 1.0$) to stabilize coefficient estimation. The resulting coefficients were exponentiated to obtain odds ratios describing the direction and magnitude of associations between genre categories and the likelihood of high engagement.

ANALYSIS

Descriptive Results

The final analytical sample included 3,967 AI-generated music videos. High viewership was defined using the 75th percentile of the view-count distribution. Videos with more than 4,058 views were classified as high-viewership videos, while those with 4,058 views or fewer were classified as lower-viewership videos(See Table 3).

Viewership group	Definition	n	%
High viewership	> 4,058 views	992	25.01
Lower viewership	≤ 4,058 views	2,975	74.99
Total		**3,967**	**100.00**

Table 3. Distribution of the High-Viewership Outcome

The resulting outcome was moderately imbalanced, with approximately one-quarter of videos belonging to the high-viewership group. The predictor set covered publication timing, genre, video and metadata characteristics, AI-platform references, AI-vocal characteristics, cover or artist-mimicry references, and disclosure characteristics. The dataset contained characteristics spanning multiple dimensions of AI-generated music content, including creator/channel characteristics, publication timing, music genre, video and metadata characteristics, AI-music platform references, AI-vocal and artist-mimicry indicators, and disclosure characteristics. These variables formed the candidate predictor set used to evaluate whether high-viewership AI-generated music videos could be distinguished from the remaining videos. Subsequent analyses compare the predictive performance of the classification models and examine which characteristics contributed most strongly to the prediction of high viewership.

Model Performance

The four classification models were evaluated on the held-out test set using accuracy, precision, recall, F1-score, and ROC-AUC. As shown in Table 4, predictive performance generally improved from Logistic Regression and Decision Tree to the ensemble models, Random Forest and XGBoost. XGBoost achieved the highest accuracy (0.861), precision (0.677), F1-score (0.756), and ROC-AUC (0.931)(see Figure 2). Random Forest achieved the highest recall (0.859), slightly exceeding XGBoost (0.854). Thus, while Random Forest identified a marginally larger proportion of high-viewership videos, XGBoost provided the strongest overall performance across the evaluation measures.

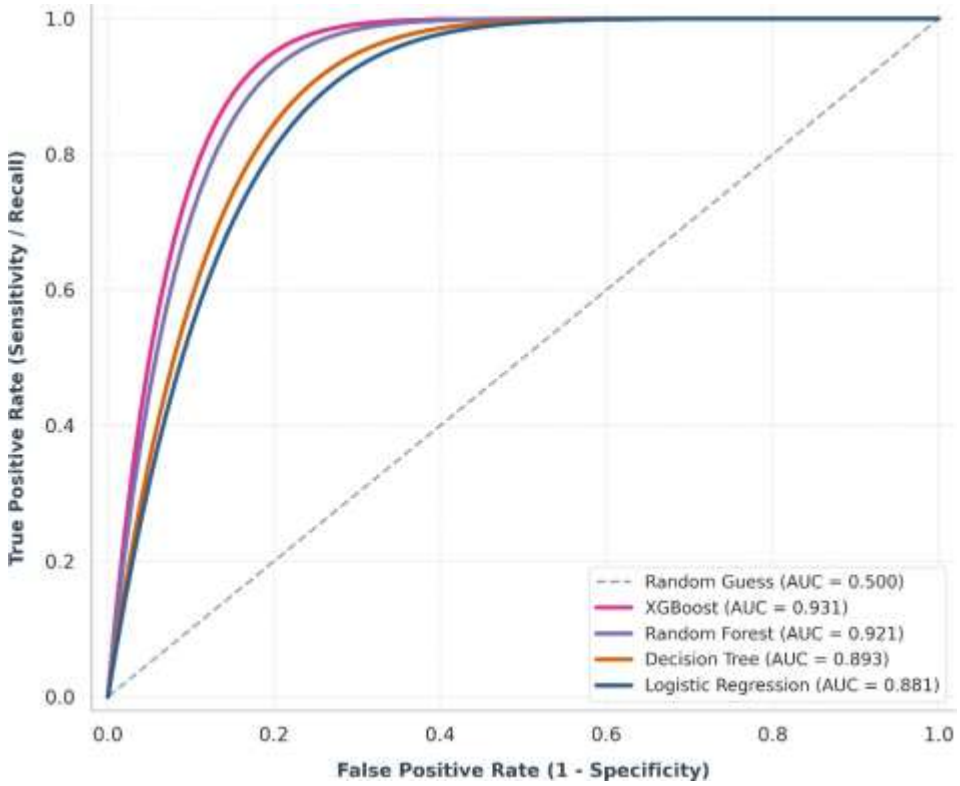


Figure 2. ROC Curve Comparison Among Classification Models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.827	0.621	0.799	0.699	0.881
Decision Tree	0.841	0.645	0.814	0.720	0.893
Random Forest	0.854	0.660	**0.859**	0.747	0.921
XGBoost	**0.861**	**0.677**	0.854	**0.756**	**0.931**

Table 4. Predictive Performance of Classification Models

The comparative pattern is also illustrated in Figure 3. The ensemble models consistently outperformed the simpler Logistic Regression and Decision Tree models across most metrics. In particular, XGBoost's ROC-AUC of 0.931 indicates strong ability to discriminate between high- and lower-viewership videos across classification thresholds. Because the positive class represented approximately one-quarter of the sample, the F1-score is particularly informative because it jointly reflects precision and recall. XGBoost achieved the highest F1-score, while maintaining high recall, supporting its selection as the primary model for the subsequent feature-importance analysis.

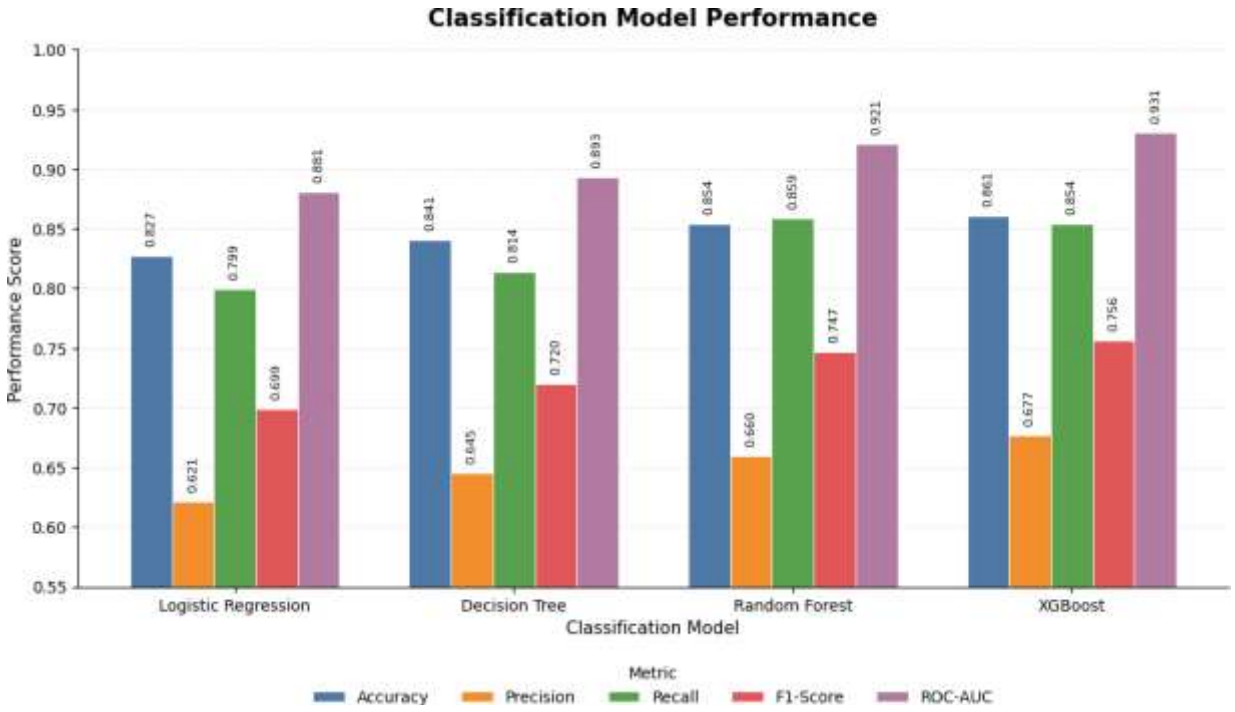


Figure 3. Predictive Performance of the Four Classification Models

FINDINGS

Predictive Feature Importance

The gain-based XGBoost feature-importance analysis indicates that genre was the strongest predictor of high viewership, with an aggregated importance score of 0.281, substantially exceeding the importance of the remaining predictors as shown in Figure 4. Cover or artist-mimicry references ranked second (0.146), followed by references to AI vocals (0.109), suggesting that both musical positioning and AI-related content characteristics played meaningful roles in distinguishing high- from lower-viewership videos. Disclosure strength showed a more moderate contribution (0.059), whereas days since upload (0.042), Suno mentions (0.039), title hashtag count (0.039), description length (0.039), total hashtag count (0.032), and Udio mentions (0.032) exhibited comparatively smaller predictive contributions. Overall, the pattern suggests that what type of AI-generated music a video contains and how it is positioned in relation to AI-enabled vocals or artist imitation are more informative for predicting high viewership than basic metadata characteristics such as hashtags, description length, or upload age. These importance values should be interpreted as measures of predictive contribution within the XGBoost model rather than as estimates of causal effects.

Genre Analysis

The genre-specific logistic regression results further clarify the direction and magnitude of the relationship between music genre and high viewership, using Classical / Jazz / Soundtracks / Instrumental as the reference category. As shown in Table 5, Several genres were associated with significantly lower odds of belonging to the high-viewership group. The strongest negative association was observed for LoFi / Ambient / Study videos (OR = 0.294, 95% CI [0.149, 0.579], p = .0004), indicating approximately 70.6% lower odds of high viewership relative to the reference category. Similarly, EDM / Electronic / Dance (OR = 0.398, p < .0001), Multi-Genre (OR = 0.404, p < .0001), R&B / Soul (OR = 0.433, p = .0071), Pop (OR = 0.479, p = .0096), Rock / Metal (OR = 0.574, p = .0200), and videos with an Unknown genre classification (OR = 0.606, p = .0030) also demonstrated statistically significant lower odds of high viewership. In contrast, Country, Covers / Artist Mimics, and HipHop / Rap were not statistically different from the reference category (p > .05). Taken together, these results indicate that although genre was the most influential predictor in the XGBoost model, its effect was not uniform across categories; rather, several specific genres were

associated with substantially lower probabilities of high viewership relative to Classical, Jazz, Soundtracks, and Instrumental content.

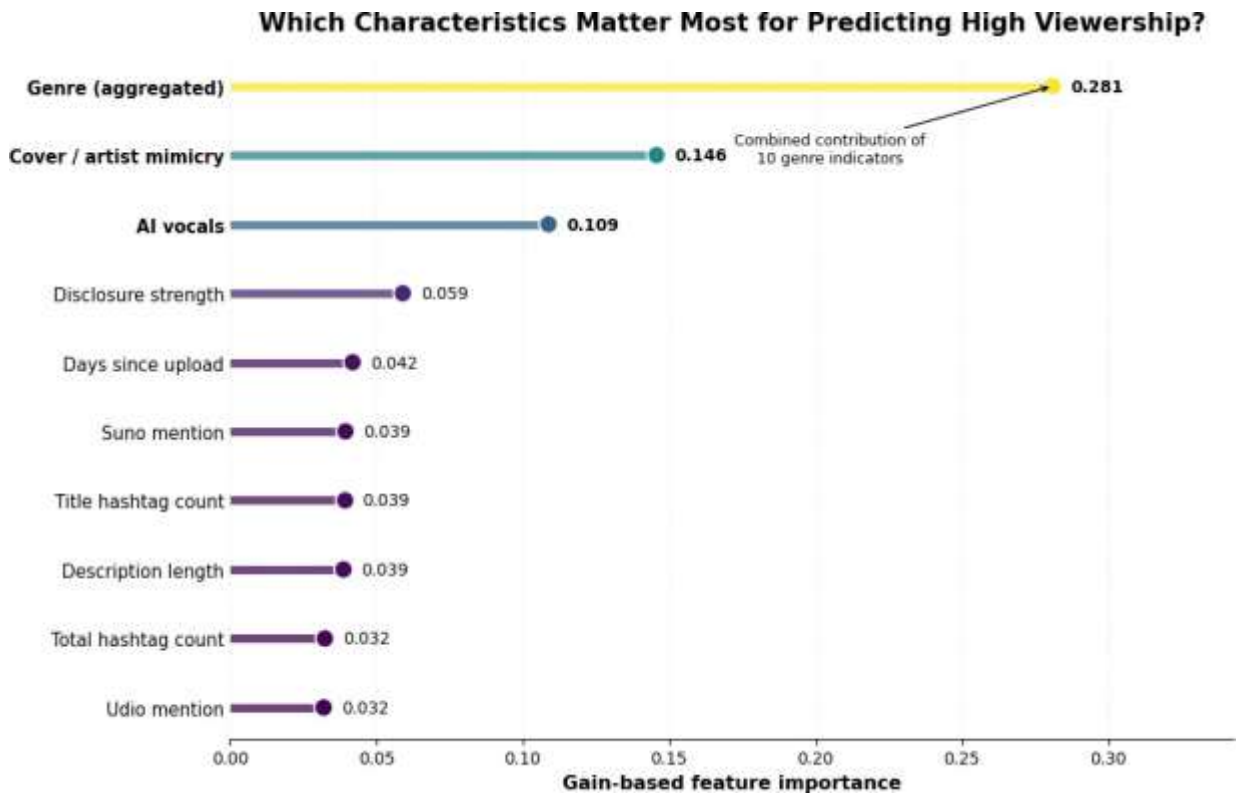


Figure 4. XGBoost Gain-Based Importance of Predictors of High Viewership

RQ1 DISCUSSION

The feature-importance analysis provides a practical indication of which characteristics are most useful for distinguishing high-viewership AI-music videos from lower-viewership videos. Genre had the greatest predictive contribution, followed by cover or artist-mimicry references and AI-vocal references, while disclosure strength, days since upload, platform mentions such as Suno and Udio, description length, and hashtag-related measures contributed less. This pattern suggests that content identity matters more than routine metadata optimization in predicting strong performance. In practice, creators may gain more from decisions about the musical genre, the use of recognizable artist-style imitation, or AI-vocal positioning than from simply adding more hashtags, changing description length, or emphasizing the specific AI platform used. For marketers and content platforms, these results indicate that recommendation, promotion, and audience-targeting systems may benefit from prioritizing content-based characteristics over relatively superficial metadata signals. Importantly, the feature set intentionally excluded raw channel-scale measures such as subscriber count and total channel views, so the findings are more informative about the characteristics of the videos themselves rather than merely reflecting the existing popularity of the channels

From a practical perspective, the genre results suggest that the type of AI-generated music being produced can meaningfully influence the likelihood of achieving high viewership. Several genres, including LoFi/Ambient/Study, EDM/Electronic/Dance, Multi-Genre, R&B/Soul, Pop, and Rock/Metal, showed significantly lower odds of high viewership compared with the Classical/Jazz/Soundtracks/Instrumental reference group. This

Genre Predictor	Coefficient (Log-Odds)	Odds Ratio	95% Confidence Interval	p-value	Direction of Effect	Statistically Significant ($p < 0.05$)?
Country	-0.221	0.802	[0.246, 2.612]	0.7143	Negative	No
Covers / Artist Mimics	-0.040	0.961	[0.617, 1.496]	0.8596	Negative	No
EDM / Electronic / Dance	-0.921	**0.398**	[0.261, 0.606]	**< 0.0001**	Negative	**Yes**
HipHop / Rap	-0.024	0.976	[0.587, 1.624]	0.9255	Negative	No
LoFi / Ambient / Study	-1.224	**0.294**	[0.149, 0.579]	**0.0004**	Negative	**Yes**
Multi-Genre	-0.906	**0.404**	[0.292, 0.558]	**< 0.0001**	Negative	**Yes**
Pop	-0.736	**0.479**	[0.275, 0.837]	**0.0096**	Negative	**Yes**
R&B / Soul	-0.837	**0.433**	[0.236, 0.797]	**0.0071**	Negative	**Yes**
Rock / Metal	-0.555	**0.574**	[0.359, 0.916]	**0.0200**	Negative	**Yes**
Unknown	-0.501	**0.606**	[0.436, 0.843]	**0.0030**	Negative	**Yes**

Note. Odds ratios below 1 indicate lower odds relative to the reference category. Bolded odds ratios, p-values, and significance indicators denote $p < 0.05$.

Table 5: Logistic Regression Odds Ratios for AI Music Genres

implies that creators, platforms, and marketers should not assume that all AI-music genres have the same audience potential. Instead, genre selection may shape discoverability, audience size, and engagement patterns. For creators, this means that content strategy should consider not only production quality but also the audience characteristics associated with different genres. For platforms or music marketers, genre can also be used as a segmentation variable when deciding which AI-generated content to promote, recommend, or target to specific audiences. At the same time, because variables such as title length, hashtags, upload timing, AI-tool mentions, and disclosure characteristics were also included in the models, the genre effects should be interpreted as associations that remain after accounting for these other observable video characteristics rather than as evidence that genre alone determines viewership.

IMPLICATIONS

The findings suggest that both platforms and creators should pay greater attention to the content characteristics of AI-generated music rather than relying mainly on superficial metadata optimization. For creators, genre choice, AI-vocal use, artist-mimicry or cover positioning, and disclosure practices may be more relevant to viewership potential than simply increasing hashtags or adjusting title length. For platforms such as YouTube, these signals could help improve recommendation, categorization, and discovery systems for AI-generated music, while clearer identification of AI-related characteristics may also support transparency and more effective audience targeting.

RQ2: EVOLUTION OF AUDIENCE THEMES AND SENTIMENT

LITERATURE REVIEW

Research on audience responses to AI music is developing through complementary methods. Fraser et al. (2021) used thematic analysis of YouTube comments on eight online music performances, identifying interaction, unity, resilience, identity, and emotion; their study establishes the value of comments for music-reception research, but did not concern AI music or temporal change. Recent cross-sectional studies examine acceptance and perception: Zhou et al. (2025) surveyed 429 music enthusiasts about using generative AI for composition, while Rigole et al. (2025) studied perceptions of AI-generated and co-created music among 57 university students and social-media users. These studies illuminate adoption intentions and listener judgments, but not how unsolicited public discourse changes over time.

Table 2.1 places the present study alongside these complementary strands. The contribution here is not a claim that no longitudinal work exists, but the combination of a large, multilingual comment corpus, AI-specific inductive themes, sentiment coding that considers translation and emoji, and quarterly analysis of change.

Study (year)	Domain / data	Methods	Key findings	Relation to current study
Fraser et al. (2021)	YouTube comments on eight online music performances during COVID-19	Online ethnography; thematic analysis	Themes included interaction, unity, resilience, identity, and emotion.	Music reception on YouTube, but not AI music or longitudinal.
Zhou et al. (2025)	Survey of 429 music enthusiasts on generative-AI composition	Questionnaire; expanded technology acceptance model	Perceived AI risk was negatively associated with use intention; content quality was positively associated.	AI-music adoption intentions; cross-sectional, not unsolicited audience comments.
Rigole et al. (2025)	University students and social-media users' perceptions of AI-generated and co-created music	Small-sample survey (n = 57)	Participants reported greater difficulty distinguishing co-created music; responses supported disclosure and openness to collaboration.	Listener perceptions, but small, cross-sectional sample and no temporal discourse analysis.
This study (2026)	YouTube comments on AI music (46,443; 12 quarters)	LLM-assisted inductive theme discovery; full-corpus coding with translation and emoji; quarterly trend tests	Realism-focused talk gives way to ethics, industry impact, and flaws; positive share of AI-related comments falls from ~66–70% to ~41–44%.	Extends audience-reception research with a multilingual, time-resolved analysis of themes and sentiment specific to AI music.

Table 2.1. Comparison of related studies and the present study.

Classical topic models such as LDA (Blei et al., 2003) and other bag-of-words approaches can miss context in short, slang-heavy, emoji-rich, and code-switched comments. Research comparing LLM annotation with crowd and expert coding (Gilardi et al., 2023; Törnberg, 2025) motivates carefully specified model-assisted analysis but does not establish that labels in this corpus are valid. We therefore report a second-model agreement check and retain human-coded validation as a requirement. The analytic workflow is informed by thematic-analysis practice (Braun & Clarke, 2006), while the themes themselves are operational codebook categories rather than a claim to reflexive thematic analysis.

Alongside this scholarly literature, contemporary announcements document changes in the environment of AI-music reception. On 18 March 2024, YouTube introduced a disclosure tool for realistic altered or synthetic content and announced a phased rollout of viewer labels, initially in the expanded description for most videos (YouTube disclosure rollout; YouTube, 2024). On 2 April 2024, the Artist Rights Alliance issued an open letter supported by more than 200 musicians concerning unauthorized AI uses and the protection of artists' work, voices, and livelihoods (Artist-rights letter; Artist Rights Alliance, 2024). Udio publicly released its text-to-music application on 10 April 2024 (Udio launch; Udio, 2024).

Rights disputes and negotiations also became prominent. Sony Music declared an opt-out from unauthorized uses of its content for AI training on 16 May 2024 (Sony training opt-out; Sony Music Group, 2024). Record companies filed copyright infringement lawsuits against Suno and Udio on 24 June 2024 concerning alleged unauthorized use of recordings in training (Suno–Udio lawsuits; Recording Industry Association of America, 2024). On 26 June 2024, the Financial Times reported negotiations between YouTube and Sony, Warner, and Universal over payments for AI-music training rights; these were reported talks rather than confirmed completed agreements (YouTube licensing talks; Nicolaou & Murgia, 2024).

Later developments emphasized authorized uses. On 16 October 2025, Spotify announced artist-first AI collaborations with Sony, Universal, Warner, Merlin, and Believe, emphasizing participation choices and compensation (Spotify AI partnerships; Spotify, 2025). On 29 October 2025, Universal Music Group and Udio announced a litigation settlement and licensing agreements for a planned authorized AI-music service (UMG–Udio agreement; Universal Music Group, 2025). These announcements and news reports establish a chronology rather than scholarly evidence of audience responses. They suggest possible contexts involving technological access, visibility of AI authorship, and rights governance. The short event names identify historical developments, not thematic categories or measured variables.

DATA AND METHODOLOGY

Data

The raw video-metadata file contained 4,002 records, representing 3,967 unique video IDs after duplicate IDs were removed. The RQ2 comment corpus included 2,344 unique videos; of these, 1,131 had at least one AI-related comment, and 1,124 contributed comments to the ten-theme analysis.The corpus contains 46,443 public comments on 2,344 YouTube videos about AI-generated music, posted between 4 July 2023 and 29 May 2026 (12 quarters; 2026 Q2 is partial). Of these, 46,408 contain text; the 35 comments without text could not be semantically coded. Emoji-only comments were retained. English accounts for 57.9% of comments, emoji-only comments 9.3%, Russian 6.9%, Hindi 4.4%, and Spanish, Romanian, Punjabi and Japanese 1.4–2.2% each. Emoji occur in 22,509 comments, of which 4,331 consist only of emoji.

Analysis Pipeline

Figure 2.1 summarizes the pipeline. (1) Theme discovery. A 4,000-comment sample (half retrieved with AI-related keywords, half random) was open-coded by gpt-4.1-mini, and gpt-4.1 consolidated the codes into a codebook of AI-specific themes (Table 2.2) with definitions and inclusion rules. Non-AI content (e.g., generic praise of a song) was explicitly excluded from the theme set. (2) Full-corpus coding. Every comment was passed with the codebook to gpt-4.1-mini (temperature 0), which returned the detected language, an English translation, an emoji reading, AI relevance (explicit, implicit, none), a primary theme and up to two secondary themes, and the sentiment toward the AI-related aspect (positive, neutral, mixed, negative). Ten very long repetitive comments were coded individually after truncation. Confidence was high for 94.9% of comments; this is model-reported confidence, not measured accuracy.

(3) Cross-model check. A stratified sample of 300 comments (100 originally labelled non-AI, 125 explicit, 75 implicit) was independently recoded with gpt-4.1; exact agreement and Cohen's κ are reported in Table 2.3. This assesses model-to-model consistency only. (4) Analysis. AI-related comments (9,773; 21.1%) were analyzed by quarter. Theme prevalence uses the primary theme (excluding 79 "other AI-related" comments, 9,694 remain over 1,124 videos); sentiment percentages per theme use all mentions (primary and secondary). Trends were tested with logistic regressions of theme membership on quarter, with video-clustered standard errors and Benjamini–Hochberg adjustment (Benjamini & Hochberg, 1995), and checked with Spearman correlations across the 12 quarters. Wilson 95% intervals are shown for quarterly shares.



Figure 2.1. Methodology and Data Flow

Theme	What the comments are about
Creation process & tools	How AI music is made: which tool, model, or prompt; tips and technical questions.
Realism & authenticity	How convincingly AI mimics real performers; doubt or surprise about whether a track is AI.
AI vs. human quality	Comparisons with human music: better, worse, lacking "soul" or emotion; preferences.
AI as creative tool	AI as an assistant or instrument for human creators; positive creative use.
Impact on artists & industry	Effects on musicians' livelihoods, jobs, labels, and the music business.
Future of AI music	Predictions and expectations about where AI music and technology are heading.
Limitations & flaws	Audible artefacts, repetitiveness, bad lyrics, and other technical shortcomings.
Ethics, consent & credit	Consent, copyright, credit, compensation, and disclosure of AI use.
Voice recreation	Emotional and ethical reactions to AI imitation of specific (often deceased) singers' voices.
AI identity & agency	Whether AI can be an artist; AI "personality," authorship, and autonomy.

Table 2.2. Codebook of AI-specific themes .

Measure	n	Exact agreement (%)	Cohen's κ
AI relevance (3 classes)	300	87.3	0.807
Primary theme (all rows)	300	74.7	0.684
Primary sentiment (all rows)	300	89.7	0.815
Primary sentiment (both models AI-related)	177	87.0	0.801

Table 2.3. ***Agreement between the original gpt-4.1-mini labels and an independent gpt-4.1 recording of the stratified 300-comment sample***

Sentiment is defined as the commenter's stance toward the AI-related aspect, not the general mood of the text; net sentiment is the share positive minus the share negative. To separate topical from tonal change, we decompose the

change in net sentiment between two periods into a *composition* effect (shift in theme mix at earlier tone) and a *within-theme* effect (change in tone at the later mix) (Table 2.4). We also compare comment-weighted estimates with equally weighted within-video shares in the first and last quarters as a sensitivity check on prolific videos (Table 2.5).

This is a descriptive, non-causal study. Historical events were examined retrospectively to contextualize the quarterly trajectories; they were not included as explanatory variables in the trend models or used to define the themes. Correspondences between event timing and discourse patterns are exploratory interpretations, and exposure to news or disclosure labels was not measured.

Theme	Share 2023 Q3–Q4 (%)	Share 2025 Q4–2026 Q2 (%)	Net 2023 Q3–Q4	Net 2025 Q4–2026 Q2	Change in net
Creation process & tools	21.7	29.4	+0.49	+0.33	-0.17
Realism & authenticity	36.6	16.2	+0.74	+0.40	-0.35
AI vs. human quality	8.2	7.9	+0.44	+0.18	-0.26
AI as creative tool	6.9	9.9	+0.97	+0.98	+0.01
Impact on artists & industry	3.5	10.3	-0.57	-0.58	-0.01
Future of AI music	5.6	6.3	+0.62	+0.55	-0.07
Limitations & flaws	4.7	10.0	-0.57	-0.77	-0.19
Ethics, consent & credit	1.5	7.0	+0.13	-0.36	-0.49
Voice recreation	11.0	2.5	+0.83	+0.50	-0.33
AI identity & agency	0.4	0.5	+0.50	+0.45	-0.05
All AI-related comments	100	100	+0.56	+0.16	-0.40
Composition effect (shift in theme mix)					-0.21 (53%)
Within-theme effect (change in tone)					-0.19 (47%)

Table 2.4. Net sentiment by theme, 2023 Q3–Q4 versus 2025 Q4–2026 Q2

Analytic frame	2023 Q3 comment-weighted (%)	2023 Q3 equal-video (%)	2026 Q2 comment-weighted (%)	2026 Q2 equal-video (%)
Wonder & authenticity	47.7	40.7	24.3	32.1
Practical use & future	36.3	44.4	38.2	39.1
Quality comparison	5.6	6.2	9.0	11.0
Critical & normative	10.3	8.6	28.5	17.8

Table 2.5. Comment-weighted and equal-video-weighted frame shares in the first and last observed quarters.

RESULTS: THE TEN THEMES

Of all comments, 21.1% were AI-related (17.6% explicit, 3.4% implicit). The AI-related share peaked at 34.6% in 2024 Q2, dropped to 15.6% in 2025 Q2, and recovered only to roughly 18–23% (Figure 2.2). Table 2.6 summarises the ten themes, ordered by number of mentions; Figure 2.3 shows quarterly shares and Figure 2.4 the overall sentiment profile. Each subsection below describes the theme, its sentiment (panel a of the figure), and its evolution (panel b). Percentages in sentiment sentences are of theme mentions; shares are of AI-related comments. OR is the odds ratio per quarter.

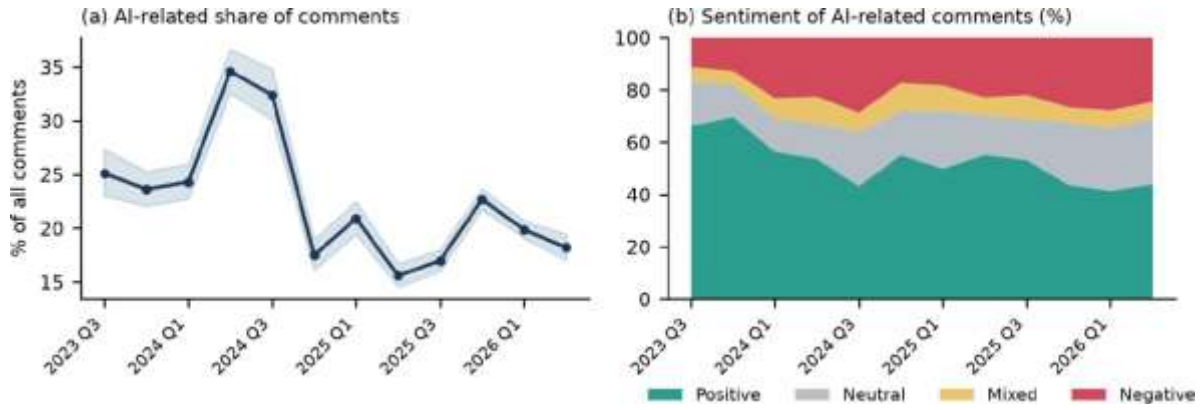


Figure 2.2. AI-Related Share of Comments and Sentiment Composition by Quarter

Theme	Ment.	Pos. %	Neu. %	Mix. %	Neg. %	Net	Overall share %	First→last qtr %	OR / qtr	p (FDR)
Creation process & tools	2,539	42.3	50.4	1.7	5.6	+0.37	24.6	24.7 → 23.3	1.06	0.013
Realism & authenticity	2,405	70.5	7.6	7.3	14.6	+0.56	21.2	41.7 → 21.4	0.91	<.001
AI vs. human quality	1,266	45.5	2.4	15.7	36.4	+0.09	10.2	5.5 → 8.2	0.97	0.099
AI as creative tool	1,172	96.2	1.3	1.5	1.0	+0.95	9.3	6.0 → 9.8	1.03	0.182
Impact on artists & industry	1,060	11.6	15.9	9.2	63.3	-0.52	8.1	3.4 → 10.6	1.08	0.014
Future of AI music	1,021	69.3	11.0	9.1	10.6	+0.59	7.2	4.9 → 4.9	0.98	0.225
Limitations & flaws	947	2.4	9.5	15.1	73.0	-0.71	8.1	6.2 → 9.9	1.07	0.001
Ethics, consent & credit	587	14.5	32.5	6.6	46.3	-0.32	4.8	0.5 → 7.8	1.13	0.006
Voice recreation	556	71.8	1.3	13.8	13.1	+0.59	4.9	5.2 → 2.8	0.85	<.001
AI identity & agency	73	47.9	38.4	6.8	6.8	+0.41	0.7	0.0 → 0.7	0.98	0.646

Table 2.6. Theme summary: sentiment, prevalence, and trend tests.

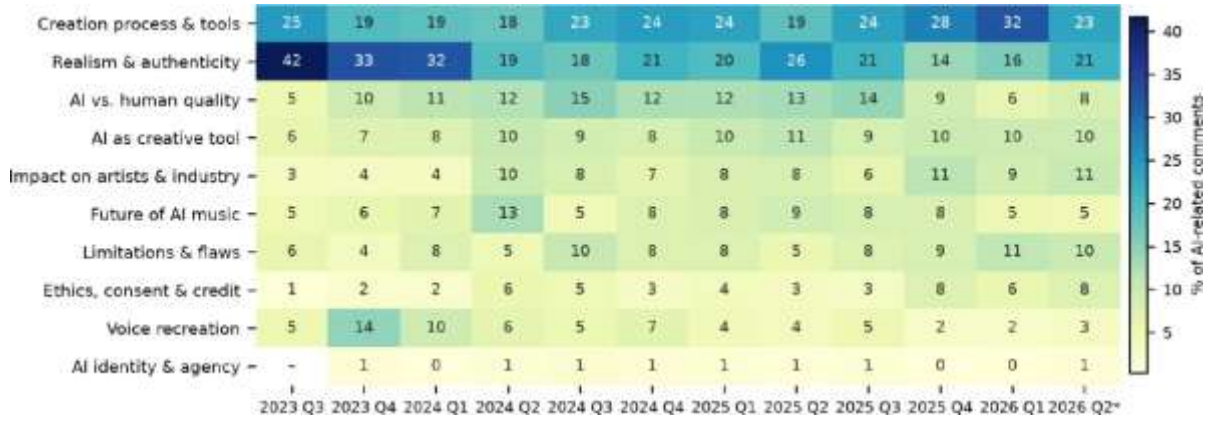


Figure 2.3. Share of AI-Related Comments by Theme and Quarter

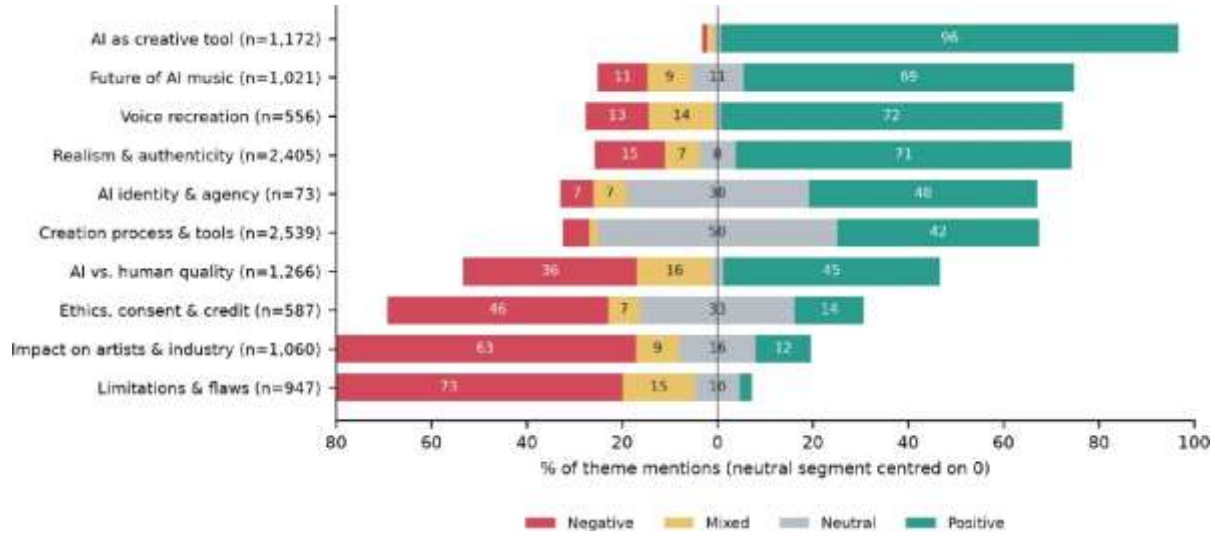


Figure 2.4. Sentiment Toward Each Theme

To assess the reproducibility of the automated coding procedure, an independent gpt-4.1 classification pass was conducted on the stratified 300-comment audit sample and compared with the original gpt-4.1-mini classifications. The two models agreed on AI relevance for 87.3% of comments, corresponding to Cohen's $\kappa = .807$. Agreement on primary theme was lower at 74.7% ($\kappa = .684$), whereas agreement on primary sentiment reached 89.7% ($\kappa = .815$). Among the 177 comments that both models classified as AI-related, sentiment agreement was 87.0% ($\kappa = .801$), as reported in Table 2.7.

Measure	n	Exact agreement (%)	Cohen's κ
AI relevance (3 classes)	300	87.3	0.807
Primary theme (all rows)	300	74.7	0.684
Primary sentiment (all rows)	300	89.7	0.815
Primary sentiment (both models AI-related)	177	87.0	0.801

Table 2.7. ***Agreement between the original gpt-4.1-mini labels and an independent gpt-4.1 recoding of the stratified 300-comment sample.***

Creation Process and Tools

The Creation Process and Tools theme captures comments focused on how AI music is produced, including the tools or models used, prompting practices, licensing considerations, and technical questions about the creation process. Typical comments included observations such as, "Riffusion doesn't grant commercial license, so it's a playtool." This was the most frequently mentioned theme and was predominantly practical and informational in nature. As shown in Figure 2.5a, sentiment within this theme was largely neutral or positive, with 42.3% of comments classified as positive, 50.4% as neutral, 1.7% as mixed, and 5.6% as negative. Over time, however, positivity declined from 52% in 2023 to 40% in 2026, while neutral sentiment increased from 43% to 51% and negative sentiment rose from 4% to 8%. This pattern suggests that discussion gradually shifted from initial enthusiasm toward more routine and matter-of-fact exchanges about AI music production.

The temporal evolution of the theme, presented in Figure 2.5b, shows that its overall share remained relatively stable, accounting for 24.7% of AI-related comments in 2023 Q3 and 23.3% in 2026 Q2. Nevertheless, there was substantial quarterly variation, including a trough of 17.7% in 2024 Q2 and a peak of 32.4% in 2026 Q1. Although the estimated trend was slightly upward (OR 1.06, $p = .013$), the relatively modest rank correlation ($\rho = .38$) indicates that this increase was non-monotonic. More broadly, the theme appears to function as a relatively stable component of AI-

related discussion: when more emotionally charged themes increased in prominence, its relative share tended to fall, whereas its share increased when those themes became less dominant.

The Udio launch provides context for the evolving production environment, while the YouTube licensing talks highlight uncertainty about authorized uses. Neither development alone explains the theme's fluctuating relative share, which also depends on the prominence of other themes.

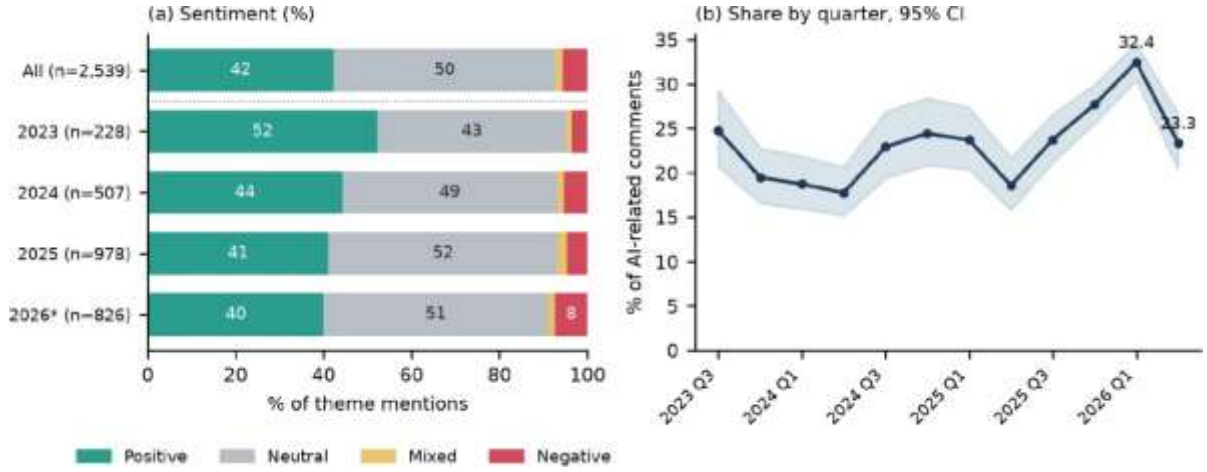


Figure 2.5. Creation Process and Tools

Realism and Authenticity

The Realism and Authenticity theme reflects reactions to the extent to which AI-generated music convincingly imitates human performance. Comments commonly expressed astonishment, disbelief, an inability to distinguish AI-generated material from human-created music or attempts to identify cues revealing its artificial origin. Examples include statements such as, "If someone told me this was a lost demo from '81 I'd believe it" and "the biggest giveaway … is the upload frequency." Sentiment toward this theme was strongly positive overall, as shown in Figure 2.6a, with 70.5% of comments classified as positive, 7.6% as neutral, 7.3% as mixed, and 14.6% as negative. However, the positive share declined substantially over time, from 83% in 2023 to 74%, 72%, and 56% in 2026. During the same period, neutral sentiment increased from 3% to 15%, while negative sentiment rose from 8% to 20%.

As illustrated in Figure 2.6b, Realism and Authenticity was the dominant theme during the early period, representing 41.7% of AI-related comments in 2023 Q3. Its share subsequently declined significantly to 21.4% in 2026 Q2, reaching a trough of 13.8% in 2025 Q4 (OR 0.91 per quarter, $p < .001$, $\rho = -.57$). This decline is consistent with diminishing novelty, although changes in the sampled videos and audiences offer alternative interpretations. The

remaining discussion shifted more toward identifying AI-generated content and questioning its authenticity, which is also consistent with the increasingly neutral and negative tone observed over time.

The YouTube disclosure rollout might also have increased attention to AI authorship and identification, providing a possible context for changing authenticity discussion; label exposure within the sampled videos was not recorded.

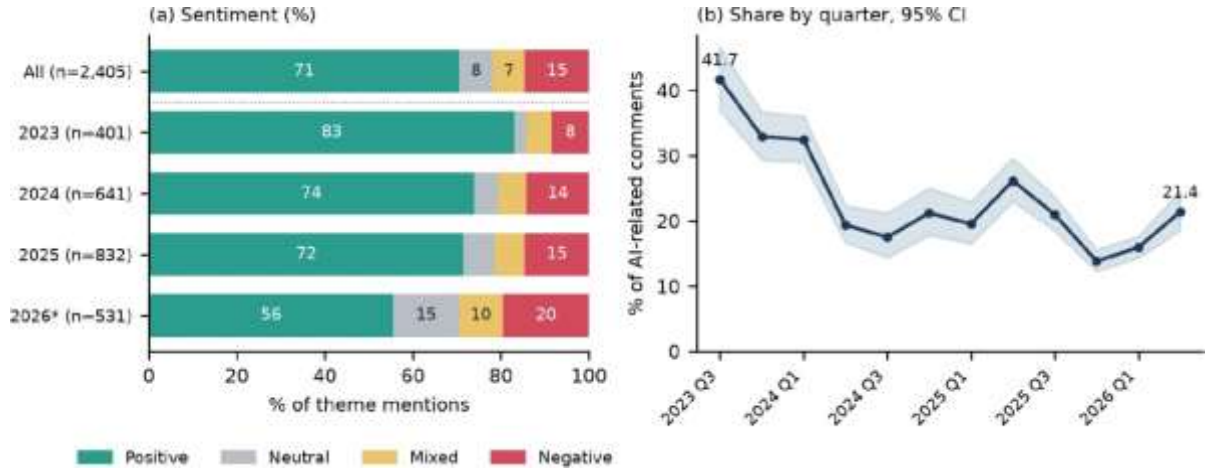


Figure 2.6. Realism and Authenticity

AI Versus Human Quality

The AI versus Human Quality theme comprises direct comparisons between AI-generated and human-created music, particularly debates over whether AI-generated outputs are equally good, superior, inferior, or lacking qualities such as emotional depth or authenticity. Comments ranged from expressions of being deceived by AI-generated content, such as “I got bamboozled by a … AI album,” to strong praise such as “This is perfection!! … AI voice.” As shown in Figure 2.7a, sentiment within this theme was highly polarised: 45.5% of comments were positive, 36.4% negative, 15.7% mixed, and only 2.4% neutral. Positive sentiment fell from 68% in 2023 to 37% in 2024 before partially recovering to 47% and 45%, while negative sentiment increased from 24% to 42% in 2024 and subsequently eased to approximately 34–36%.

The temporal pattern shown in Figure 2.7b indicates that the theme’s share increased from 5.5% to 8.2%, although this change did not constitute a statistically significant trend (OR 0.97, p = .099). Discussion peaked at 15.0% in 2024 Q3 before declining to 6.4% in 2026 Q1. This pattern suggests a relatively concentrated period of intense debate over whether AI-generated music could rival or surpass human-created music, followed by a reduction in the prominence of such direct comparisons.

The Udio launch preceded this peak and offers a plausible context for heightened comparison of AI and human capabilities. Whether the peak reflects continuing attention to that release would require finer temporal and video-level evidence.

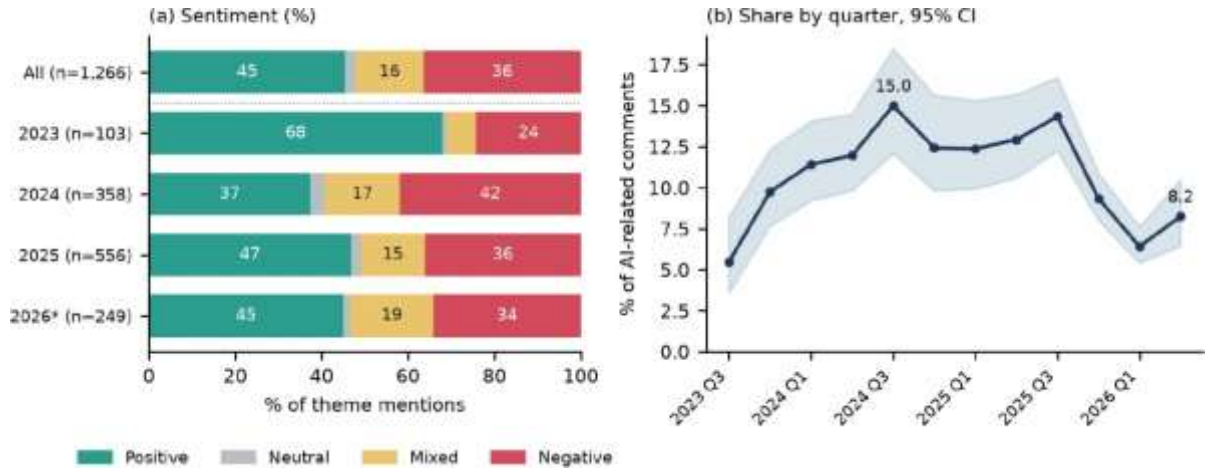


Figure 2.7. AI Versus Human Quality

AI as a Creative Tool

The AI as Creative Tool theme captures comments in which AI is framed as an instrument that supports or extends human creativity rather than replacing it. This perspective was frequently expressed by creators describing their own workflows or emphasizing the role of human skill in determining the quality of AI-assisted output, as illustrated by the statement, "AI is just another tool, talent is how you use it." Sentiment toward this theme was overwhelmingly positive, as shown in Figure 2.8a, with 96.2% of comments classified as positive and only 1.0% as negative. This pattern remained highly stable across years, with positive sentiment ranging between 95–97%. However, because the theme itself is defined by a favourable framing of AI as a creative aid, the near-ceiling level of positivity should be interpreted partly as a consequence of the thematic definition rather than as evidence of general acceptance of AI-generated music.

As shown in Figure 2.8b, the theme's share increased from 6.0% to 9.8%, reaching a peak of 11.1%, although the estimated trend was not statistically significant (OR 1.03, p = .18). These results indicate that the theme represented a consistent and supportive minority perspective throughout the study period. Its relative prominence increased modestly, but this increase appears to have resulted partly from declines in other themes rather than from a substantial independent expansion of this viewpoint.

The Spotify AI partnerships and UMG–Udio agreement illustrate the coexistence of creative-use opportunities and rights governance. This context is consistent with the supportive tool-oriented perspective, without explaining its prevalence or implying broad audience acceptance.

Impact on Artists and the Industry

The Impact on Artists and the Industry theme encompasses discussion of the consequences of AI-generated music for musicians' livelihoods, record labels, creative labor, and the broader platform ecosystem. Some comments expressed concern about the increasing volume of AI-generated material, as reflected in the phrase "The slop era of YouTube," while others praised human-created music in explicit contrast to AI-generated content, for example by stating that a work was "made this without AI." A smaller subset of comments defended AI as a potential source of income. Sentiment within this theme was predominantly negative, as shown in Figure 2.9a, with 63.3% of comments classified as negative, 11.6% as positive, 15.9% as neutral, and 9.2% as mixed. Negative sentiment remained consistently high across the study period, accounting for 60%, 56%, 65%, and 66% of comments by year. Importantly, positive sentiment in this theme does not necessarily imply approval of AI: some positive comments praised human creators

or discussed income opportunities associated with AI. Approximately 17% of a spot-checked sample matched human-made keywords, further indicating that positive sentiment should be interpreted cautiously.

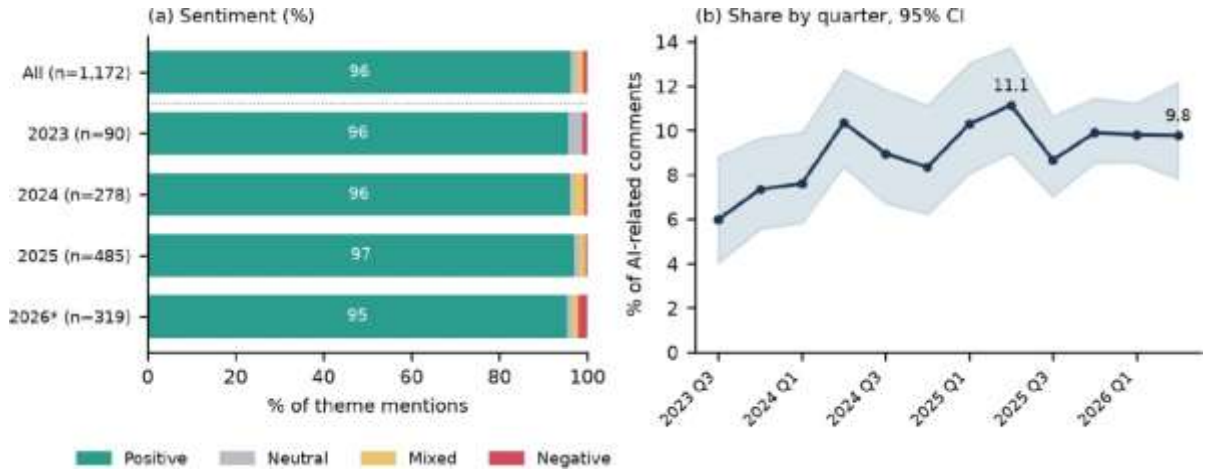


Figure 2.8. AI as a Creative Tool

The temporal evolution presented in Figure 2.9b shows a substantial increase in the theme's prominence. Its share tripled from 3.4% to 10.6%, with a notable step increase to 9.9% in 2024 Q2 and a peak of 11.3% in 2025 Q4 (OR 1.08, $p = .014$, $\rho = .71$). The sentiment distribution changed relatively little over this period; instead, the primary change was the increasing volume of discussion. Concerns regarding artists' livelihoods and the broader industry therefore developed from a comparatively minor topic into a recurring and sustained component of AI-related discourse.

The Artist-rights letter and Sony training opt-out offer a plausible context for increased attention to artists' livelihoods in the earlier spike. The Suno–Udio lawsuits and YouTube licensing talks added to this debate only near the quarter's end. The later peak coincided with the Spotify AI partnerships and UMG–Udio agreement, when authorized uses and compensation were prominent industry issues.

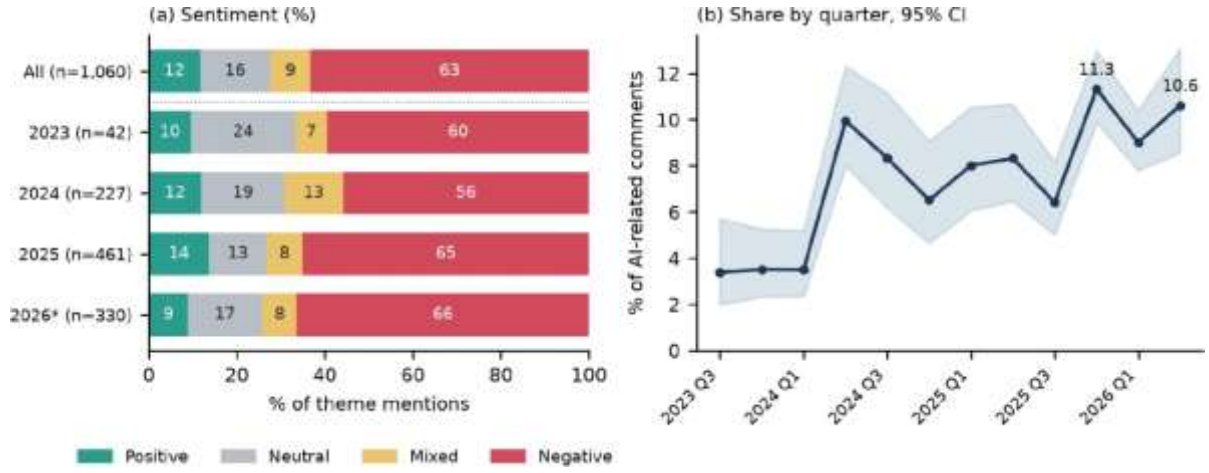


Figure 2.9. Impact on Artists and the Industry

Future of AI Music

The Future of AI Music theme includes speculation about the direction of AI-generated music, including expectations regarding future quality, wider adoption, and the potential implications for human-created music. As shown in Figure 2.10a, sentiment was predominantly positive, with 69.3% of comments classified as positive, 11.0% as neutral, 9.1% as mixed, and 10.6% as negative. Positive sentiment remained relatively stable across years, ranging from 69–73%, although negativity increased modestly in 2026 to 14%. Notably, the most-liked comments within this theme were more negative than the theme-level sentiment distribution as a whole, suggesting that highly visible reactions were not necessarily representative of the broader discussion.

The temporal distribution shown in Figure 2.10b indicates that the theme was broadly stable, representing 4.9% of AI-related comments at both the beginning and end of the study period (OR 0.98, p = .22). The principal exception was a spike to 12.6% in 2024 Q2, coinciding with the broader peak in AI-related discussion during that period. The Udio launch offers a plausible context for the spike by bringing new music-generation capabilities into public view.

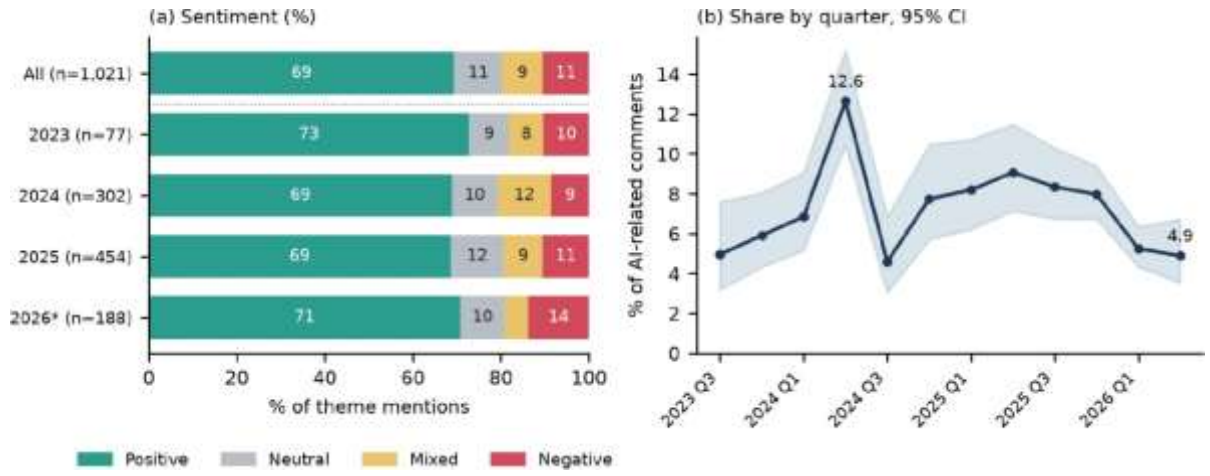


Figure 2.10. Future of AI Music

Limitations and Flaws

The Limitations and Flaws theme encompasses criticism of audible or structural shortcomings in AI-generated music, including repetitive composition, robotic vocals, weak lyrics, and the broader characterization of low-quality output as "AI slop." A representative comment stated, "This song is so pretty, but the AI is so noticeable." As shown in Figure 2.11a, this was the most negative theme apart from ethics, with 73.0% of comments classified as negative, 15.1% as mixed, 9.5% as neutral, and only 2.4% as positive. Negative sentiment also intensified over time, increasing from 61% in 2023 to 71%, 72%, and 77% in 2026.

The temporal pattern in Figure 2.11b shows that the theme's share increased from 6.2% to 9.9%, reaching a peak of 11.1% in 2026 Q1 (OR 1.07, p = .001, ρ = .69). A supplementary lexical check provides additional context for this change: the term "slop" was absent in 2023 but appeared in 1.0% of AI-related comments in 2026. This shift suggests that criticism evolved from references to specific audible deficiencies toward a more general label used to characterize mass-produced AI-generated content.

Ethics, Consent, and Credit

The Ethics, Consent and Credit theme captures concerns relating to consent, copyright, attribution, compensation, misleading upload practices, the use of individuals' likenesses, and potentially bot-like praise of AI-generated uploads. Sentiment within this theme was mixed but predominantly negative. As shown in Figure 2.12a, 46.3% of comments were negative, 32.5% neutral, 14.5% positive, and 6.6% mixed. The tone became substantially more negative over time: positive sentiment declined from 32% in 2023 to 13% in 2025–2026, while negative sentiment increased from

26% to 50%, 47%, and 45%. These early-year estimates should be interpreted cautiously because 2023 n = 19, making the corresponding percentages relatively imprecise.

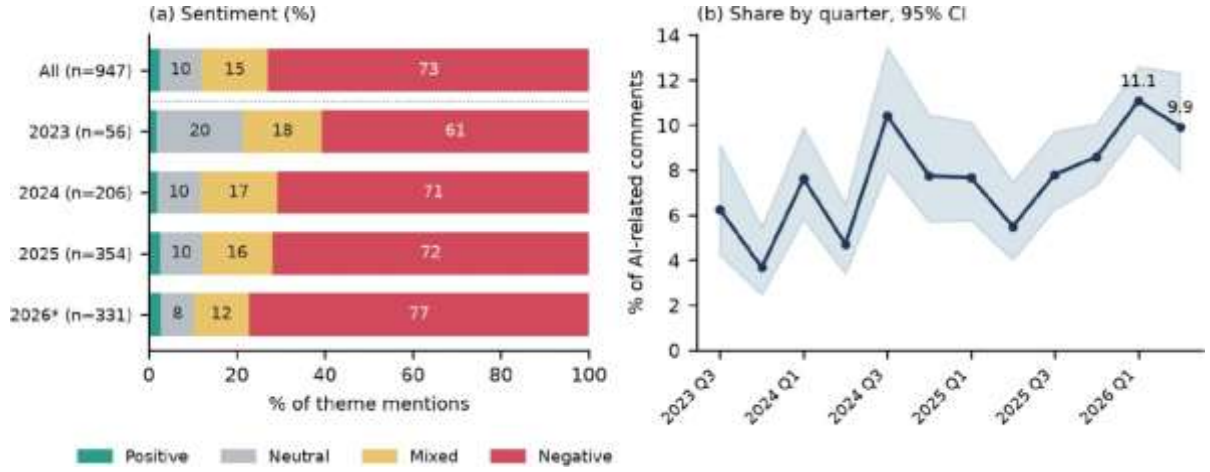


Figure 2.11. Limitations and Flaws

As shown in Figure 2.12b, Ethics, Consent and Credit was the fastest-growing theme in the analysis. Its share increased from 0.5% of AI-related comments in 2023 Q3 to 7.8% in 2026 Q2 (OR 1.13, p = .006, ρ = .78). The pattern included a step increase to 5.5% in 2024 Q2, followed by a decline to approximately 3% across 2024 Q4–2025 Q3 and a renewed rise to 7.7% in 2025 Q4. Consistent with this thematic expansion, references to copyright, consent, and related terms increased from 1.3% to 4.1% of AI-related comments. Together, these patterns indicate that ethical and rights-based concerns became increasingly prominent within discussions of AI-generated music.

The YouTube disclosure rollout, Artist-rights letter, and Sony training opt-out may help explain greater attention to transparency and permissions during the earlier increase. The Suno–Udio lawsuits and YouTube licensing talks provide additional late-quarter context, while the Spotify AI partnerships and UMG–Udio agreement contextualize the renewed attention to licensing and compensation in the later increase.

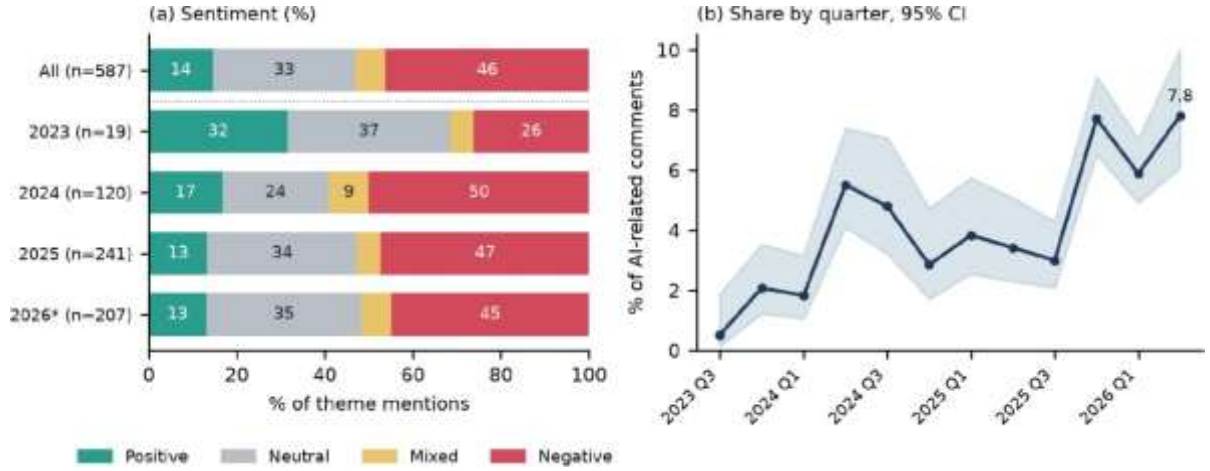


Figure 2.12. Ethics, Consent, and Credit

Voice Recreation

The Voice Recreation theme captures emotional and ethical reactions to AI-generated reproductions of recognizable singers' voices, particularly those of deceased artists such as Michael Jackson. One representative comment stated, "Even after 37 years of his death the world still seeks that voice." Overall sentiment was positive, as shown in Figure 2.13a, with 71.8% of comments classified as positive, 13.8% as mixed, 13.1% as negative, and 1.3% as neutral. However, positivity declined steadily across the study period, from 87% in 2023 to 71%, 66%, and 62%, while mixed sentiment increased from 8% to 19% and negative sentiment rose from 3% to 16–17%. Interpretation of negative sentiment requires caution because comments expressing grief about deceased singers may be classified as negative even when they do not represent disapproval of AI voice recreation itself.

The temporal pattern in Figure 2.13b shows that Voice Recreation peaked at 14.4% in 2023 Q4 before declining to 2.8% in 2026 Q2 (OR 0.85, $p < .001$, $\rho = -.78$), representing the steepest decline among the identified themes. This pattern is consistent with diminished novelty, although changing video and audience composition may also contribute. The Artist-rights letter offers a plausible context for greater attention to voice imitation and consent; the concurrent rise in ethics discussion does not establish that individual commenters moved between themes.

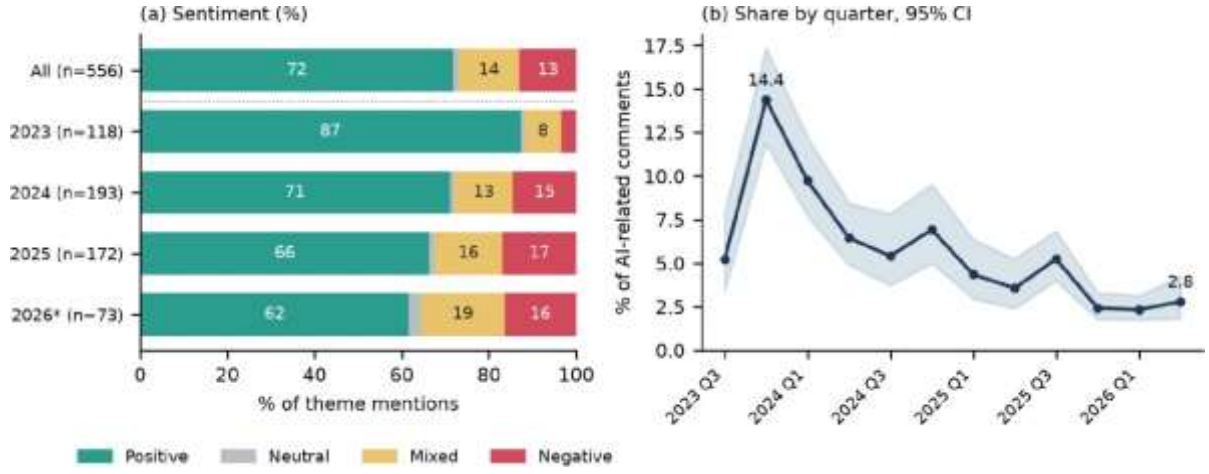


Figure 2.13. Voice Recreation

AI Identity and Agency

The AI Identity and Agency theme addresses questions about whether AI can meaningfully be considered an "artist," possess an artistic persona, or deserve authorship or creative recognition. Comments included statements such as "Self-proclaimed AI 'artists' …" and reflected differing views regarding the role and status of AI in creative production. As shown in Figure 2.14a, the theme was relatively small and sentiment was mixed, with 47.9% of comments classified as positive, 38.4% as neutral, 6.8% as mixed, and 6.8% as negative. Negative sentiment reached 18% in 2026 (n = 17), although the small number of observations limits the reliability of year-specific interpretation.

The temporal analysis in Figure 2.14b shows that the theme remained below 1.5% of AI-related comments throughout the study period and exhibited no statistically significant trend (OR 0.98, $p = .65$). Given the total of only 73 mentions, no further substantive interpretation is made.

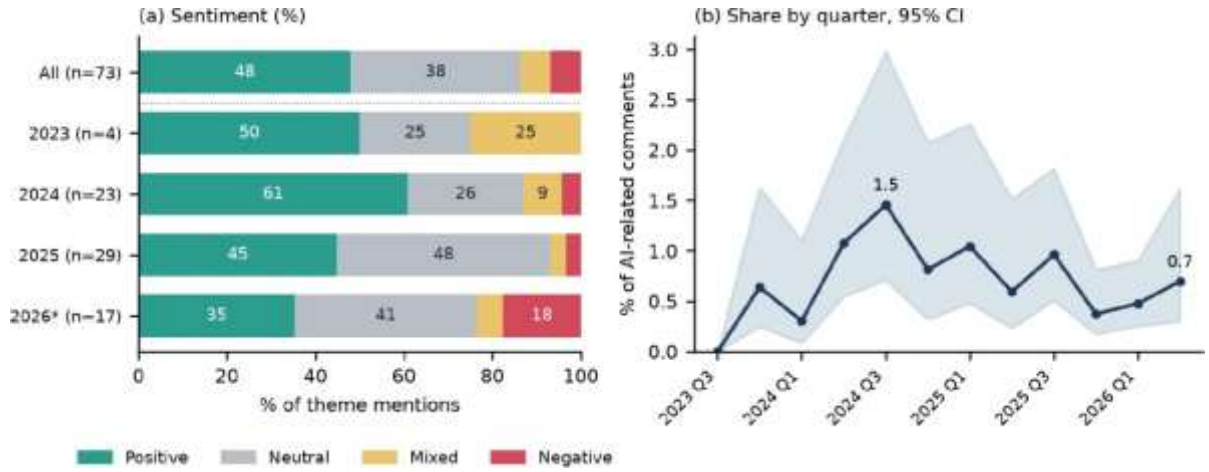


Figure 2.14. AI Identity and Agency

FINDINGS: CROSS-THEME

From Wonder to Critique

Read together, the themes show a reorientation from perception to consequence. Grouping them into four frames (Figure 2.15), the wonder-and-authenticity frame (realism, voice recreation) accounted for 47.6% of AI-related comments in 2023, then 30.4%, 22.4%, and 20.1% in 2024–2026. The critical-and-normative frame (impact on artists, ethics, limitations) rose from 9.7% to 18.3%, 22.2%, and 26.9%, overtaking wonder by 2025 Q4. By quarter, the first fell from 47.7% to 24.3% and the second rose from 10.3% to 28.5%. Early audiences asked "is this real?"; later audiences asked "who is harmed, who is credited, and is this any good?" Figure 2.16 shows the same exchange at theme level. Under equal-video weighting, wonder still falls (40.7% to 32.1%) and critique still rises (8.6% to 17.8%), although both shifts are smaller (Table 2.8). The Spotify AI partnerships and UMG–Udio agreement provide context for the later prominence of governance concerns. This reorientation describes sampled discourse rather than a demonstrated change in the views of the same individuals.

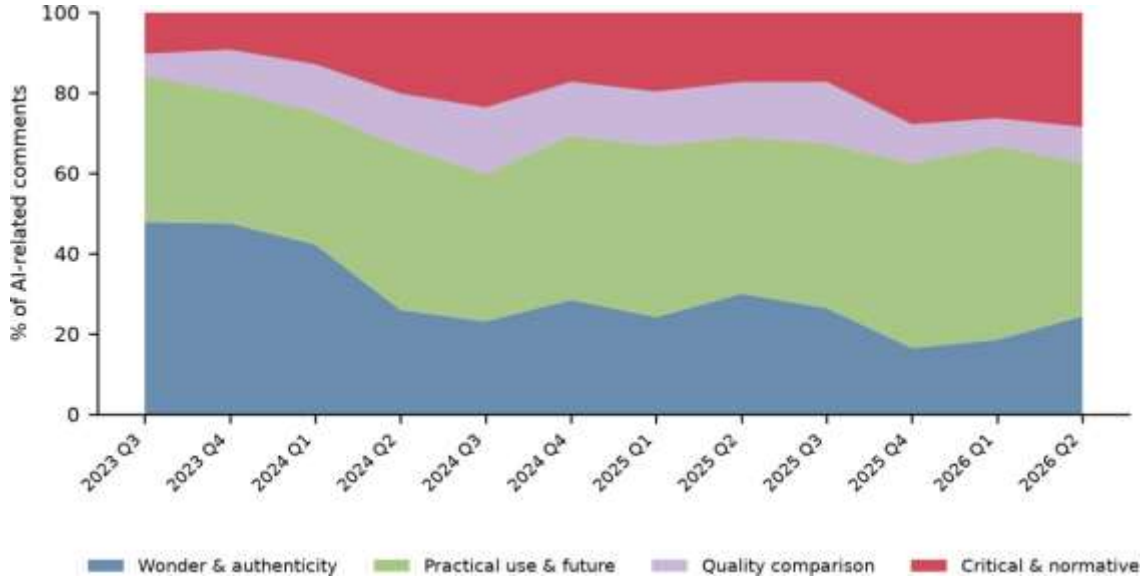


Figure 2.15. Analytic Frames Over Time

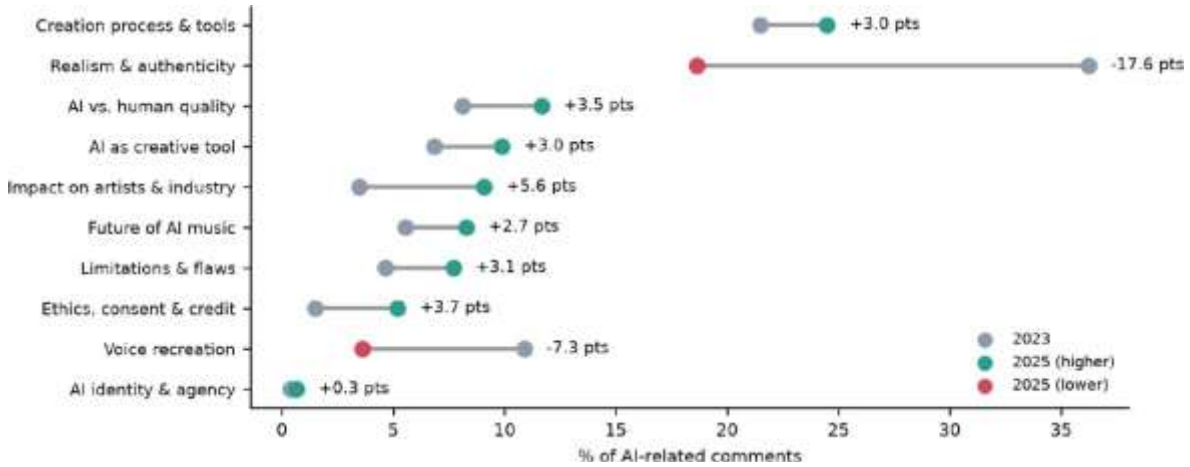


Figure 2.16. Change in theme share between 2023 and 2025 (percentage points of AI-related comments).

Analytic frame	2023 Q3 comment-weighted (%)	2023 Q3 equal-video (%)	2026 Q2 comment-weighted (%)	2026 Q2 equal-video (%)
Wonder & authenticity	47.7	40.7	24.3	32.1
Practical use & future	36.3	44.4	38.2	39.1
Quality comparison	5.6	6.2	9.0	11.0
Critical & normative	10.3	8.6	28.5	17.8

Table 2.8. Comment-weighted and equal-video-weighted frame shares in the first and last observed quarters.

A Stable Practical Core

Creation process and tools, AI as a creative tool, and the future of AI music together account for a substantial practical frame. In comment-weighted data it rose from 34.2% to 37.9%, 43.0%, and 45.4% by year, but the equal-video comparison does not show the same increase: practical use & future declines from 44.4% to 39.1% between the two endpoint quarters (Table 2.9). Thus the practical frame remains sizeable, but its apparent growth is sensitive to weighting and video mix. The positive tool-oriented community persists alongside a growing critical audience; these are coexisting strands rather than a simple replacement. The Spotify AI partnerships and UMG–Udio agreement similarly illustrate how creative opportunities and rights concerns can coexist in the wider industry environment.

Cooling Sentiment: Changing Topics, Changing Tone

The positive share of AI-related comments was 66.1% and 69.6% in 2023 Q3–Q4 and 41–44% in 2025 Q4–2026 Q2, while negative rose from 11.2% to 24.6% and neutral from 17% to 26% (Figure 2.2b). Net sentiment fell from +0.56 to +0.16 (Table 2.9). Our decomposition attributes about −0.21 (53%) to composition—the move from highly positive themes (realism, voice) to negative ones (impact, limitations, ethics)—and about −0.19 (47%) to within-theme cooling. The within-theme part is most visible for realism (net +0.74 to +0.40), voice recreation (+0.83 to +0.50), and above all ethics (+0.13 to −0.36); impact (−0.57 to −0.58) and creative tool (+0.97 to +0.98) were unchanged (Figure 2.17). The overall cooling therefore reflects both changing topical composition and less positive aggregate tone within several themes. Within-theme change does not establish individual attitude change; the rights-related developments discussed below offer context for the greater salience of critical discussion.

Theme	Share 2023 Q3–Q4 (%)	Share 2025 Q4–2026 Q2 (%)	Net 2023 Q3–Q4	Net 2025 Q4–2026 Q2	Change in net
Creation process & tools	21.7	29.4	+0.49	+0.33	-0.17
Realism & authenticity	36.6	16.2	+0.74	+0.40	-0.35
AI vs. human quality	8.2	7.9	+0.44	+0.18	-0.26
AI as creative tool	6.9	9.9	+0.97	+0.98	+0.01
Impact on artists & industry	3.5	10.3	-0.57	-0.58	-0.01
Future of AI music	5.6	6.3	+0.62	+0.55	-0.07
Limitations & flaws	4.7	10.0	-0.57	-0.77	-0.19
Ethics, consent & credit	1.5	7.0	+0.13	-0.36	-0.49
Voice recreation	11.0	2.5	+0.83	+0.50	-0.33
AI identity & agency	0.4	0.5	+0.50	+0.45	-0.05
All AI-related comments	100	100	+0.56	+0.16	-0.40
Composition effect (shift in theme mix)					-0.21 (53%)
Within-theme effect (change in tone)					-0.19 (47%)

Table 2.9. Net sentiment by theme, 2023 Q3–Q4 versus 2025 Q4–2026 Q2

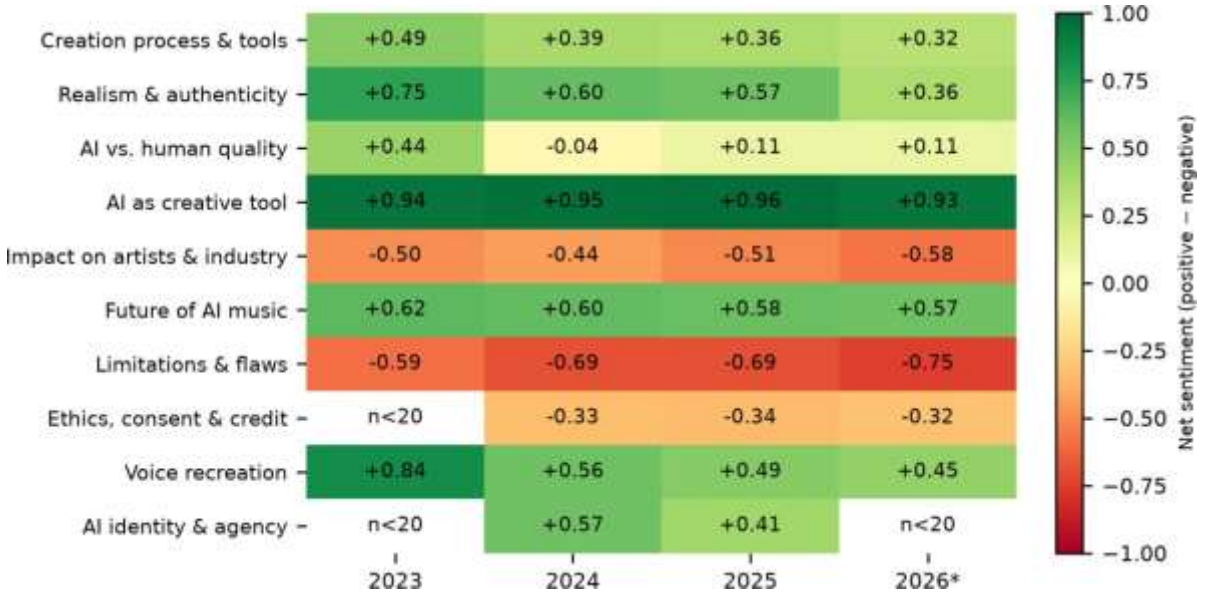


Figure 2.17. Net Sentiment by Theme and Year

Normative and Economic Vocabulary

Ethics, impact, and limitations rise together, and the vocabulary of the corpus supports a shift toward normative judgement. Copyright, consent, permission, royalty, and theft terms rose from 1.3% to 4.1% of AI-related comments, and "slop" from 0.0% to 1.0%, whereas "soul" remained around 1.1–1.6% and real/fake terms about 11–12%. The authenticity vocabulary, in other words, did not vanish; it persisted at constant volume while new evaluative language was added. Within the "wonder" period, voice recreation was often tender (fans of deceased singers). Its subsequent decline alongside increased rights vocabulary is consistent with greater attention to consent, contextualized by the Artist-rights letter and Sony training opt-out, without demonstrating that the same conversations migrated between themes.

Event-Like Spikes and Volatility

The AI-related share, the future-of-AI-music theme, ethics, and impact all show a step or spike in 2024 Q2, and AI vs. human quality peaks the following quarter. The pattern is consistent with a period of intense, heterogeneous discussion followed by greater prominence of ethics and impact. The YouTube disclosure rollout may have made AI authorship more visible, while the Udio launch offered new capabilities to discuss. The Artist-rights letter and Sony training opt-out provide a plausible context for greater attention to permissions and livelihoods. The Suno–Udio lawsuits and YouTube licensing talks added legal conflict and licensing debate near the end of the quarter; they cannot explain comments posted before those developments. The following quarter's quality-comparison peak may reflect continuing attention to AI capabilities, but this possible sequence requires finer temporal evidence and assessment of the sampled video mix.

The later rise in ethics and industry-impact discussion coincided with the Spotify AI partnerships and UMG–Udio agreement. These developments may help explain increased attention to the conditions under which AI music is produced and commercialized. The shift toward critical-and-normative discussion can therefore be read in the context of consent, compensation, and governance, rather than as uniform rejection of AI. Changes in sampled videos and audiences remain alternative explanations, and event-specific references in comments warrant follow-up.

RQ2 DISCUSSION

Using full-corpus, LLM-assisted coding of 46,443 multilingual YouTube comments, we find that audience discussion of AI-generated music moved from astonishment at realism and voice cloning to a more critical, normative, and economic conversation, with a stable positive core of practical, tool-focused discussion. Sentiment cooled from net +0.56 to about +0.16, partly because the topics changed and partly because tone within topics, especially ethics, worsened. The study provides an AI-specific theme set and a reproducible, time-resolved picture of audience reception. The observed reorientation coincided with technological releases, rights disputes, and licensing developments, suggesting that audience reception should be interpreted within its changing institutional environment.

6. Implications

For platforms and rights-holders, the growth of ethics and impact themes, and their consistently negative tone, highlights the relevance of disclosure, consent mechanisms for voice and likeness, and concerns about mass-uploaded "slop." For creators, the stable positive tool-oriented community suggests receptiveness to AI as an assistant when authorship is clear. The YouTube disclosure rollout, Spotify AI partnerships, and UMG–Udio agreement illustrate responses to these concerns, but the findings do not establish which interventions would improve reception.

CONCLUSION

Considered together, the two analyses reveal an important distinction between what attracts attention to AI-generated music and what audiences increasingly discuss once engaging with it. RQ1 shows that content-related characteristics—particularly genre, artist mimicry or cover positioning, and AI-vocal references—are more informative for predicting high viewership than routine metadata characteristics. RQ2, however, shows that audience discourse surrounding AI-generated music has gradually shifted away from initial fascination with realism and voice recreation toward questions involving artists, ethics, consent, credit, and technological limitations. Thus, characteristics associated with the visibility of AI music do not necessarily correspond to the issues that increasingly shape audience evaluation of that content.

One particularly notable connection concerns imitation and AI-generated voices. RQ1 identifies cover or artist-mimicry references and AI-vocal references among the strongest predictors of high viewership. At the same time, RQ2 demonstrates that realism, voice recreation, artist impact, and ethics form important components of audience discourse. Together, these findings suggest a potential tension in the AI-music ecosystem: characteristics that make AI-generated music distinctive and potentially attractive to audiences may also be closely connected to the authenticity, consent, attribution, and artist-impact concerns that increasingly characterize public discussion

LIMITATIONS AND FUTURE RESEARCH

This study has several limitations that should be considered when interpreting the findings as a whole. Because the analysis is based only on YouTube, the results may not fully represent AI-generated music on other platforms, where audiences and recommendation systems may behave differently. In addition, some important factors could not be observed, such as paid promotion, recommendation-system exposure, creator reputation, audience characteristics, and direct exposure to AI labels or outside news. The study also relies mainly on video metadata and coded comment content rather than the actual audio, lyrics, thumbnails, or visual features. Most importantly, the findings show relationships and changes over time, but they do not prove that particular video characteristics cause higher viewership or that specific industry events cause changes in audience attitudes. Future research could address these limits by following the same videos and audiences over longer periods, combining video characteristics directly with their comment themes and sentiment, using adding audio and visual features, and comparing platforms such as YouTube, Spotify, and TikTok. This would make it possible to examine more directly whether the same characteristics that help AI-generated music attract attention also influence how audiences react to it over time.

REFERENCES

Agostinelli, A., Denk, T. I., Borsos, Z., Engel, J., Verzetti, M., Caillon, A., Huang, Q., Jansen, A., Roberts, A., Tagliasacchi, M., Sharifi, M., Zeghidour, N., & Frank, C. (2023). *MusicLM: Generating music from text*. arXiv. https://doi.org/10.48550/arXiv.2301.11325

Artist Rights Alliance. (2024, April 2). *200+ artists call on AI developers, tech platforms not to devalue music and undermine artists' rights* [Press release]. https://thetrichordist.com/wp-content/uploads/2024/04/araopenletterai-release.pdf

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological), 57*(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research, 3*, 993–1022.

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology, 3*(2), 77–101. https://doi.org/10.1191/1478088706qp063oa

Copet, J., Kreuk, F., Gat, I., Remez, T., Kant, D., Synnaeve, G., Adi, Y., & Défossez, A. (2023). Simple and controllable music generation. *Advances in Neural Information Processing Systems, 36*. https://doi.org/10.52202/075280-2066

Fraser, T., Crooke, A. H. D., & Davidson, J. W. (2021). "Music has no borders": An exploratory study of audience engagement with YouTube music broadcasts during COVID-19 lockdown, 2020. *Frontiers in Psychology, 12*, Article 643893. https://doi.org/10.3389/fpsyg.2021.643893

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). *ChatGPT outperforms crowd-workers for text-annotation tasks*. arXiv. https://arxiv.org/abs/2303.15056

Lee, J., & Lee, J.-S. (2018). Music popularity: Metrics, characteristics, and audio-based prediction. *IEEE Transactions on Multimedia, 20*(11), 3173–3182. https://doi.org/10.1109/TMM.2018.2820903

Martín-Gutiérrez, D., Hernández Peñaloza, G., Belmonte-Hernández, A., & Álvarez-García, F. (2020). A multimodal end-to-end deep learning architecture for music popularity prediction. *IEEE Access, 8*, 39361–39374. https://doi.org/10.1109/ACCESS.2020.2976033

Nicolaou, A., & Murgia, M. (2024, June 26). YouTube in talks with record labels over AI music deal. *Financial Times*. https://www.ft.com/content/e2d9472d-32e0-43f5-8109-efb753fac330

Recording Industry Association of America. (2024, June 24). *Record companies bring landmark cases for responsible AI against Suno and Udio in Boston and New York federal courts, respectively* [Press release].

https://www.riaa.com/record-companies-bring-landmark-cases-for-responsible-ai-againstsuno-and-udio-in-boston-and-new-york-federal-courts-respectively/
Reisz, N., Servedio, V. D. P., & Thurner, S. (2024). Quantifying the impact of homophily and influencer networks on song popularity prediction. *Scientific Reports, 14*(1), 8929. https://doi.org/10.1038/s41598-024-58969-w
Rigole, N. J., Sandoval, Z. V., Beasley, S., & Lingelbach, K. (2025). Perceptions of AI-generated and co-created music among university students and social media users. *Issues in Information Systems, 26*(4), 102–117. https://doi.org/10.48009/4iis2025110
Saragih, H. S. (2023). Predicting song popularity based on Spotify's audio features: Insights from the Indonesian streaming users. *Journal of Management Analytics, 10*(4), 693–709. https://doi.org/10.1080/23270012.2023.2239824
Sony Music Group. (2024, May 16). *Declaration of AI training opt out*. https://www.sonymusic.com/sonymusic/declaration-of-ai-training-opt-out/
Spotify. (2025, October 16). *Sony Music Group, Universal Music Group, Warner Music Group, Merlin, and Believe to partner with Spotify to develop artist-first AI music products*. https://newsroom.spotify.com/2025-10-16/artist-first-ai-music-spotify-collaboration/
Terroso-Saenz, F., Soto, J., & Muñoz, A. (2023). Music mobility patterns: How songs propagate around the world through Spotify. *Pattern Recognition, 143*, 109807. https://doi.org/10.1016/j.patcog.2023.109807
Törnberg, P. (2025). Large language models outperform expert coders and supervised classifiers at annotating political social media messages. *Social Science Computer Review, 43*(6), 1181–1195. https://doi.org/10.1177/08944393241286471
Udio. (2024, April 10). *Former Google DeepMind researchers assemble luminaries across music and tech to launch Udio, a new AI-powered app that allows anyone to create extraordinary music in an instant* [Press release]. PR Newswire. https://www.prnewswire.com/news-releases/former-google-deepmind-researchers-assemble-luminaries-across-music-and-tech-to-launch-udio-a-new-ai-powered-app-that-allows-anyone-to-create-extraordinary-music-in-an-instant-302113071.html
Universal Music Group. (2025, October 29). *Universal Music Group and Udio announce Udio's first strategic agreements for new licensed AI music creation platform* [Press release]. https://www.universalmusic.com/universal-music-group-and-udio-announce-udios-first-strategic-agreements-for-new-licensed-ai-music-creation-platform/
Weatherbed, J. (2026, May 27). YouTube is putting AI labels where you'll actually see them. *The Verge*. https://www.theverge.com/streaming/937915/youtube-ai-labels-shorts-automatic-identification-updates
Yee, Y. K., & Raheem, M. (2022). Predicting music popularity using Spotify and YouTube features. *Indian Journal of Science and Technology, 15*(36), 1786–1799. https://doi.org/10.17485/IJST/v15i36.2332
YouTube. (2024, March 18). *How we're helping creators disclose altered or synthetic content*. https://blog.youtube/news-and-events/disclosing-ai-generated-content/
Zhou, Y., Wang, Y., & Su, J. (2025). A study on factors influencing music enthusiast's behavioral intention to use generative AI for music composition based on an expanded technology acceptance model. *Scientific Reports, 15*, Article 44802. https://doi.org/10.1038/s41598-025-28532-2

CONTACT INFORMATION

Mohsen Pashna: Mpashna@okstate.edu

Hunter Willard: hunter.willard@okstate.edu